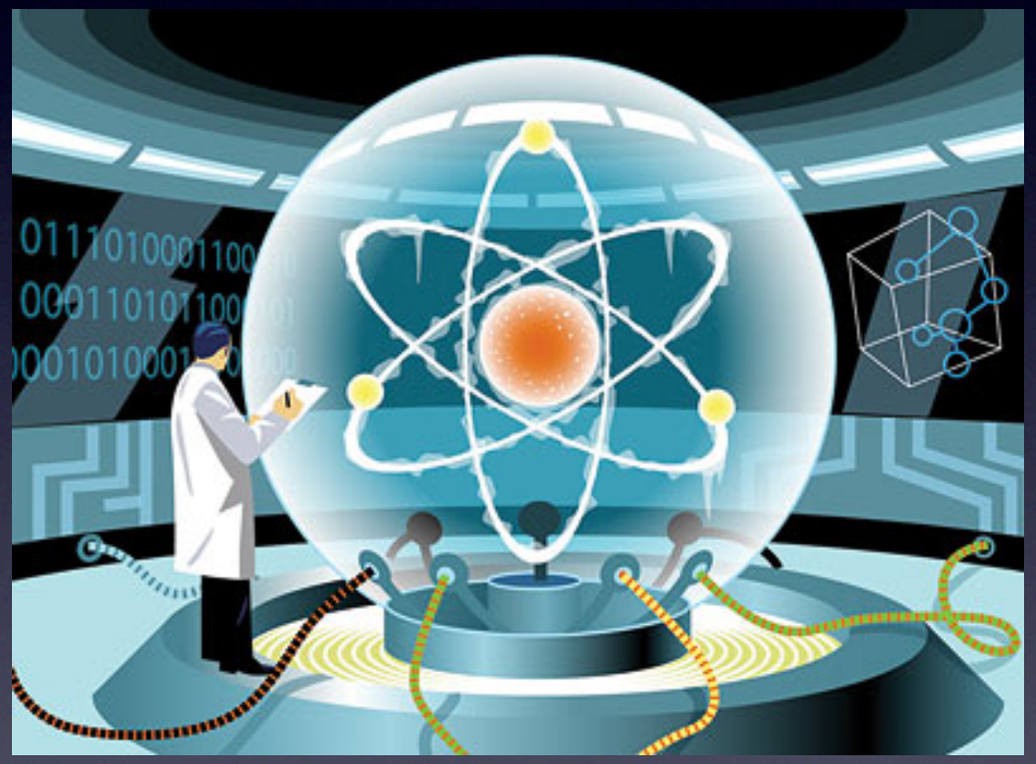
Test Slide



Video Games on Quantum Computers



Dave Bacon (dabacon@cs.washington.edu)
University of Washington

Video Games on Quantum Computers?????



Dave Bacon (dabacon@cs.washington.edu)
University of Washington

Nothing New Under the Sun



!!!!!

Rumors

Dissecting the Project Cafe Controller Rumors

Exploring the potential te ontroller.

April 15, 2011



Paul Tassi

INSERT COIN

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intendo's Secret
Controller?

Apr. 21 2011 - 9:43 am | *60 views* | *0 recommendations* | *0 comments*

GAMES BLOG

Previous | Blog home

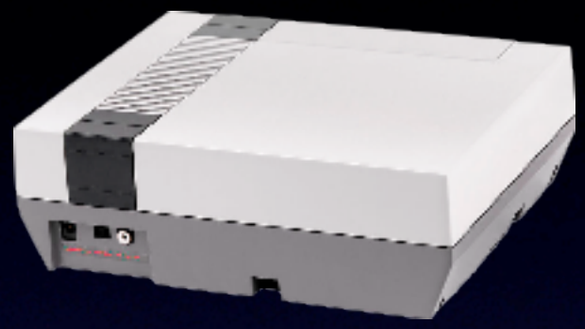
Wii 2: all the rumours in one handy story

An E3 unveiling for Nintendo's Wii successor is now almost certain as industry gossip factory kicks into gear.

sa Bacon

They're All Wrong

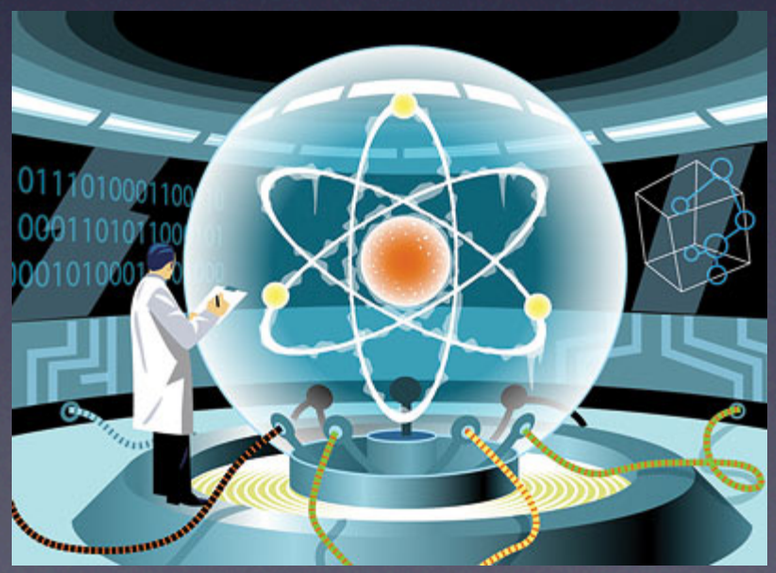








Nintendo's next system is a quantum computer!*

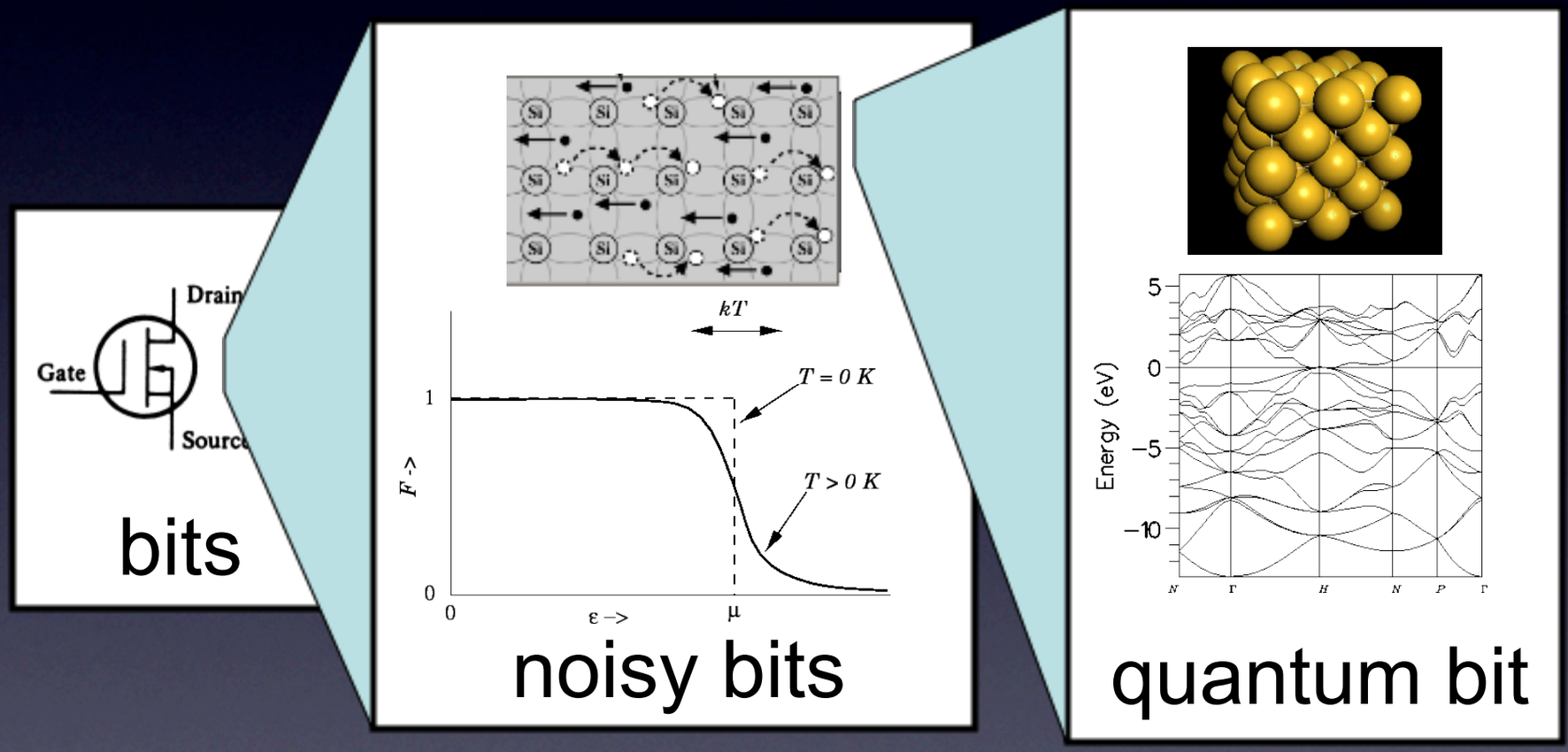






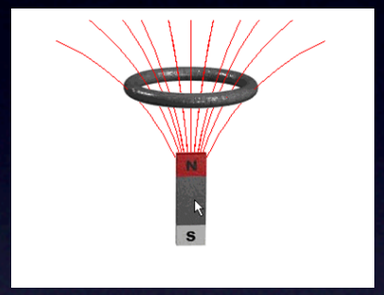
* non-factual

Under the Hood

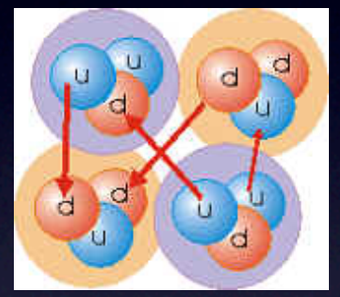


What is Quantum Theory?

Electromagnetism

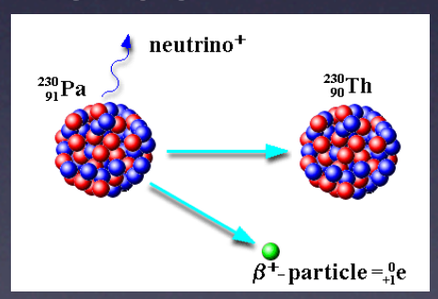


Strong force

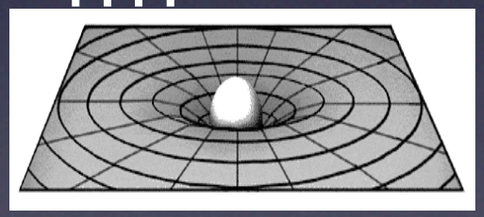




Weak force



Gravity ?!?!



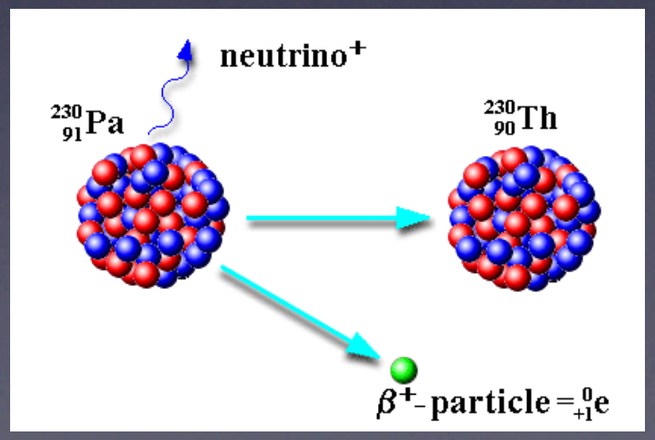
Quantum Theory

Special Relativity

Where is Quantum Theory?

Quantum **Classical**

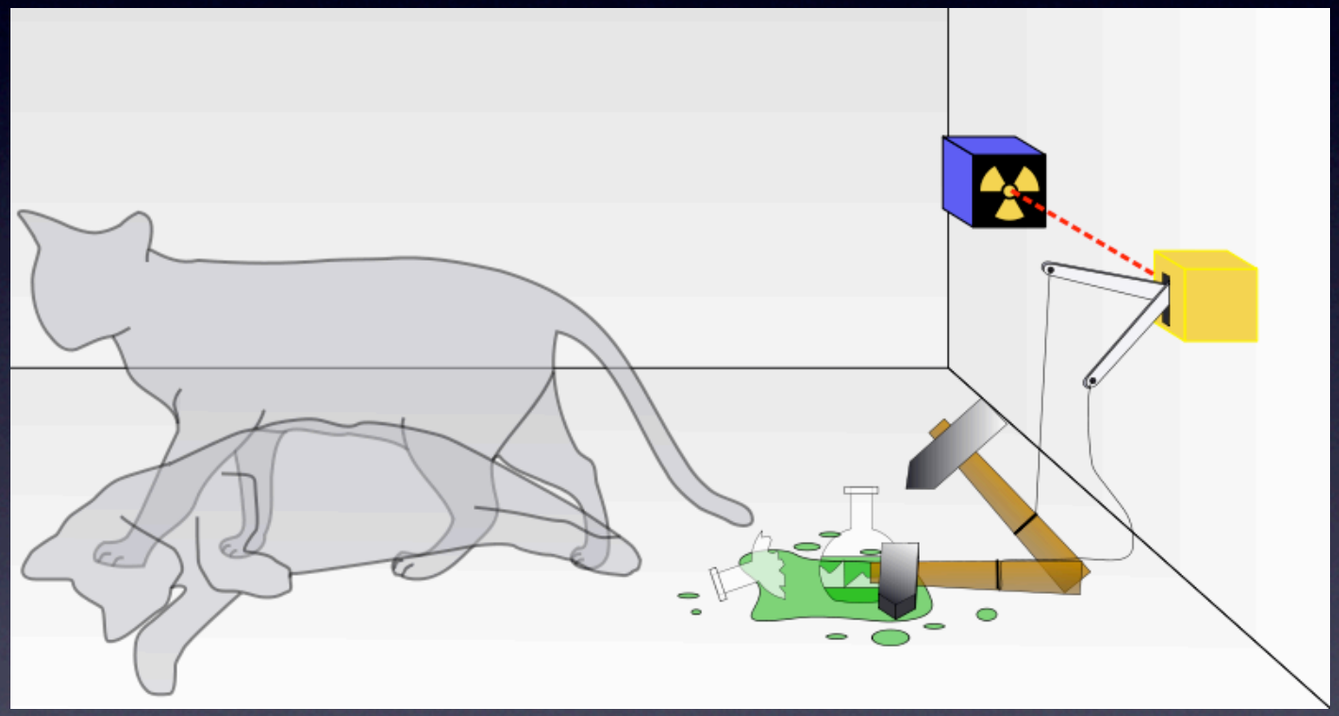
small “size” large

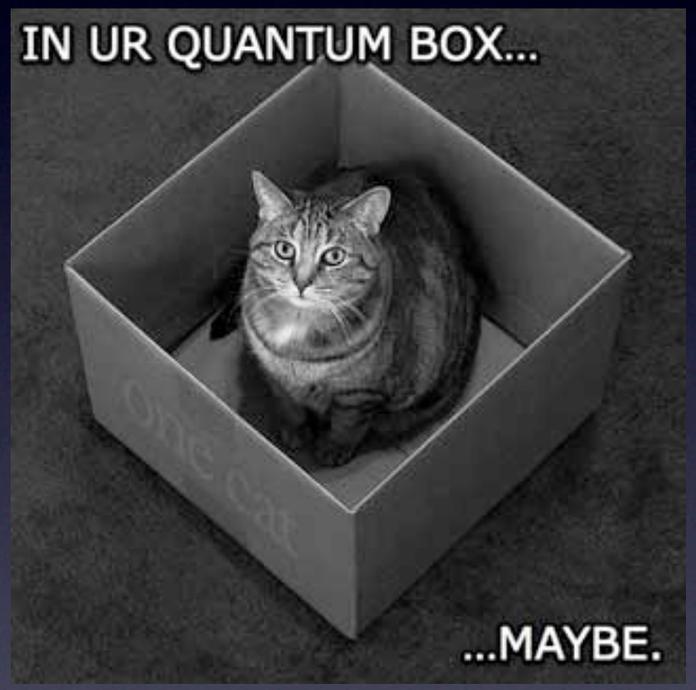




Quantum Weirdness

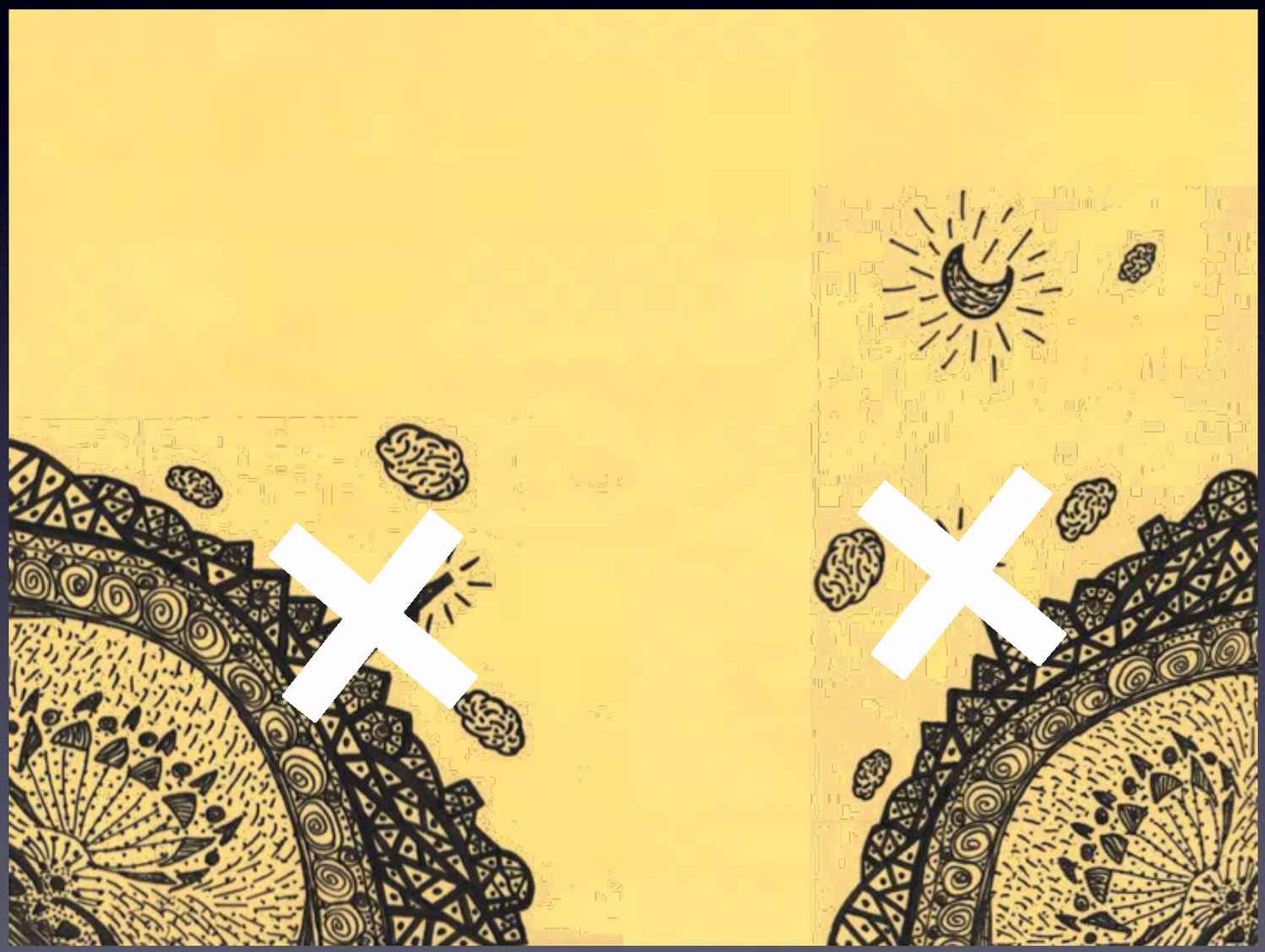
Cats that are Alive and Dead?





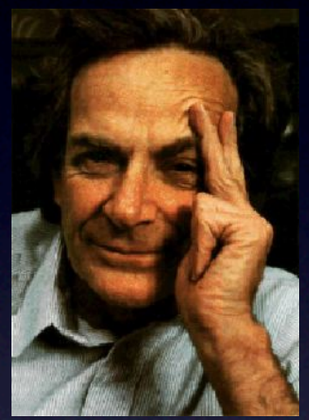
Quantum Weirdness

Nonlocality?



Einstein's "Spooky action at a distance"

Quantum Propaganda



I think I can safely say
that nobody
understands quantum
mechanics.

Richard Feynman
Nobel Prize 1965



Anyone who is
not shocked by quantum
theory has not understood it.

Niels Bohr
Nobel Prize 1922

Hogwash

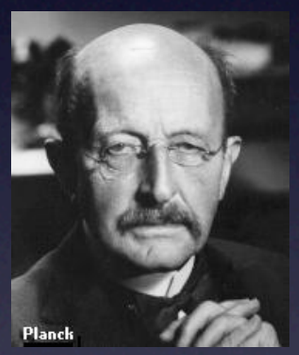


- They understand it
- They use it
- But do they believe it?

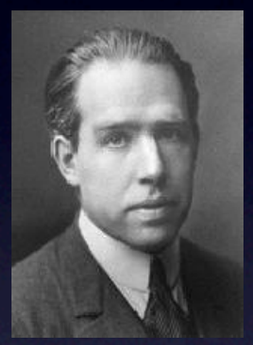
Quantum History

Inventing Worrying Accepting!

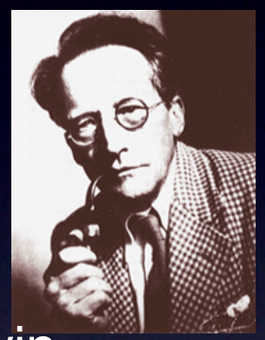
1900 1910 1920 1930 1940 1950 1960 1970 1980 1990 2000



Max Planck



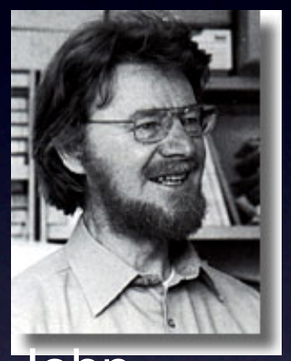
Niels Bohr



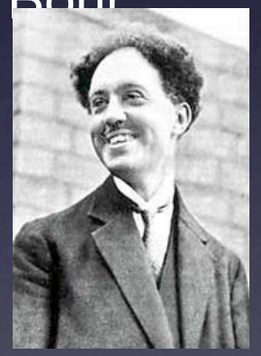
Erwin Schrodinger



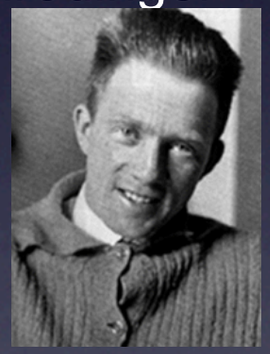
John Von Neumann



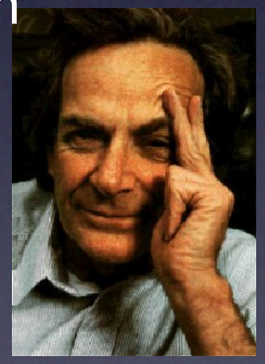
John Bell



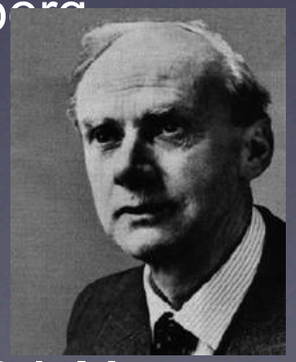
Louis de Broglie



Werner Heisenberg

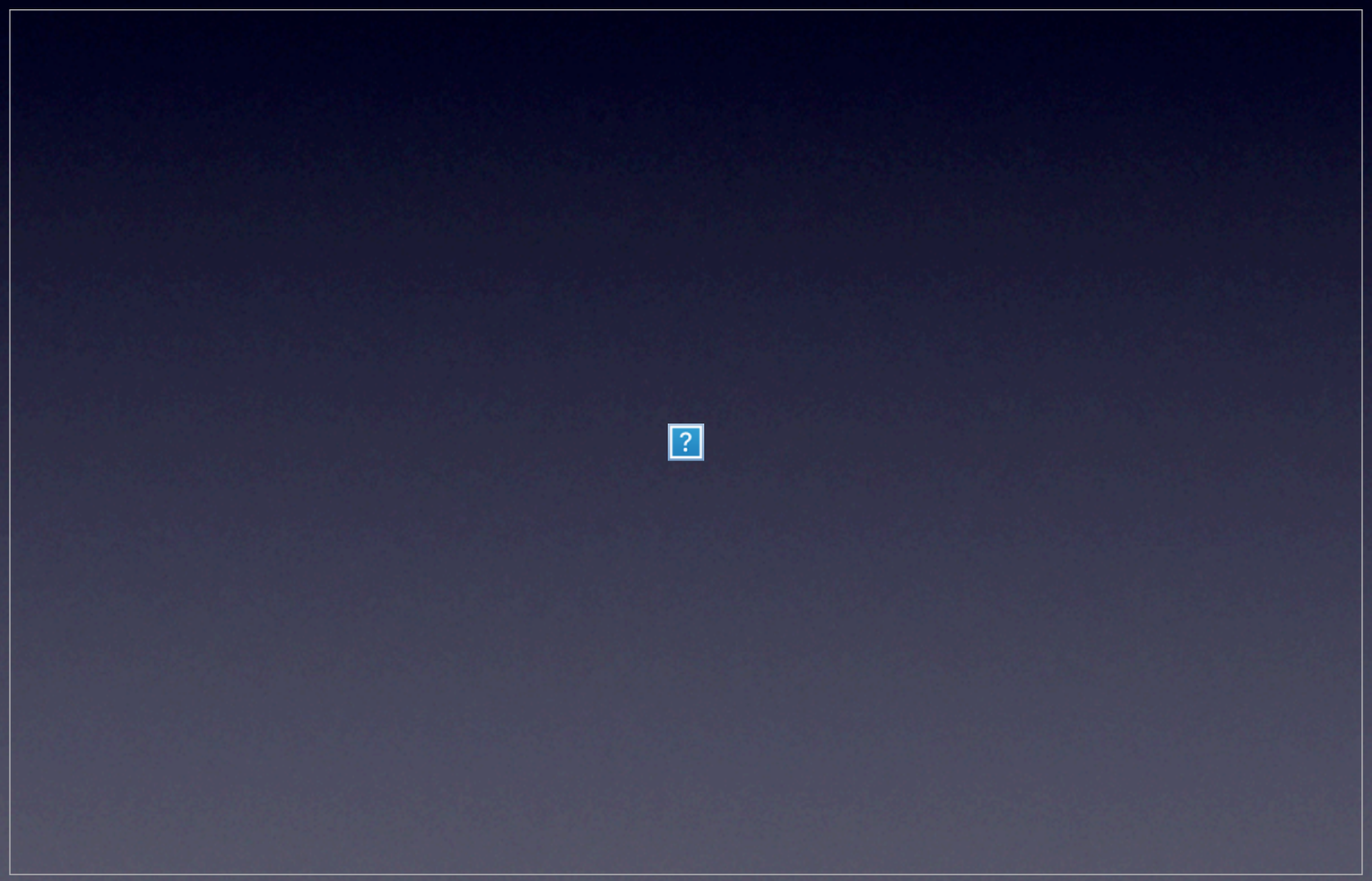


Richard Feynman



P.A.M. Dirac

Stop Worrying and Love Quantum Theory



Richard Feynman:

"Do not ask yourself, if you can possibly avoid that, 'how can it be like that?' because you will lead yourself down a blind alley in which no one has ever escaped."

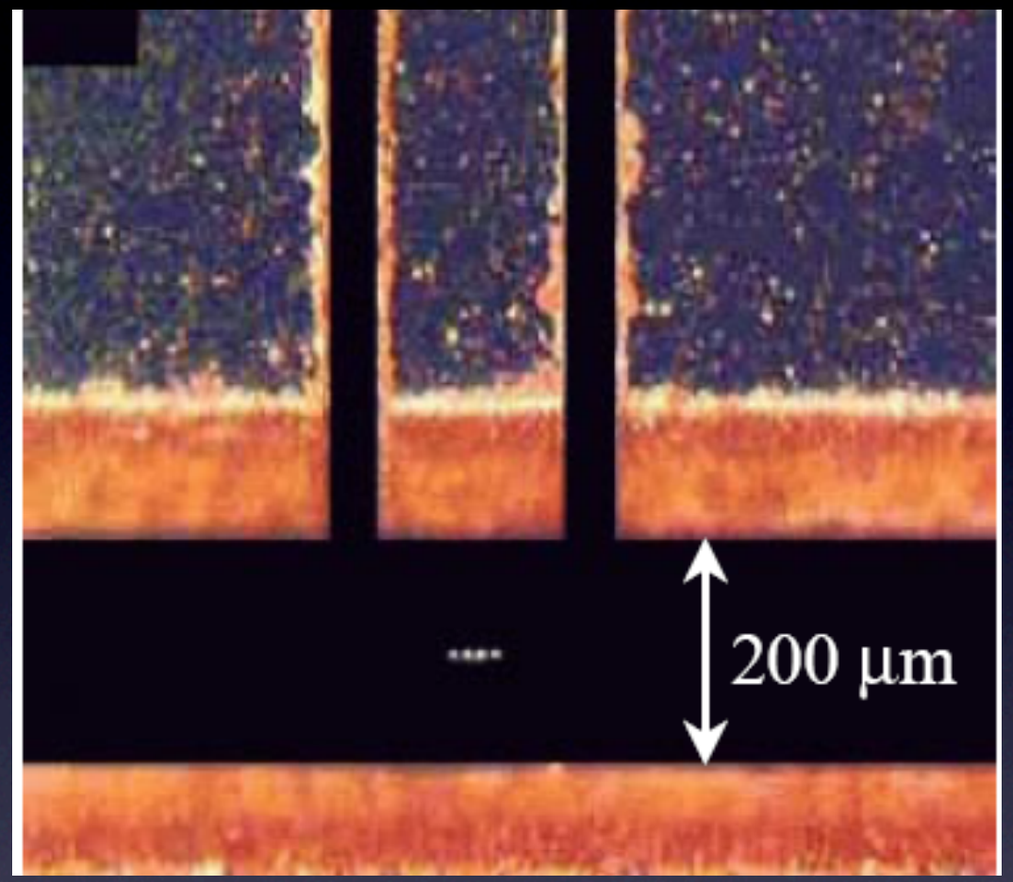
"For a successful technology, nature must take precedence over public relations, for nature cannot be fooled."

"How can we simulate quantum mechanics?...Can we do it with a new kind of computer…a quantum computer? It's not a Turing machine but a machine of a different kind" (1981)

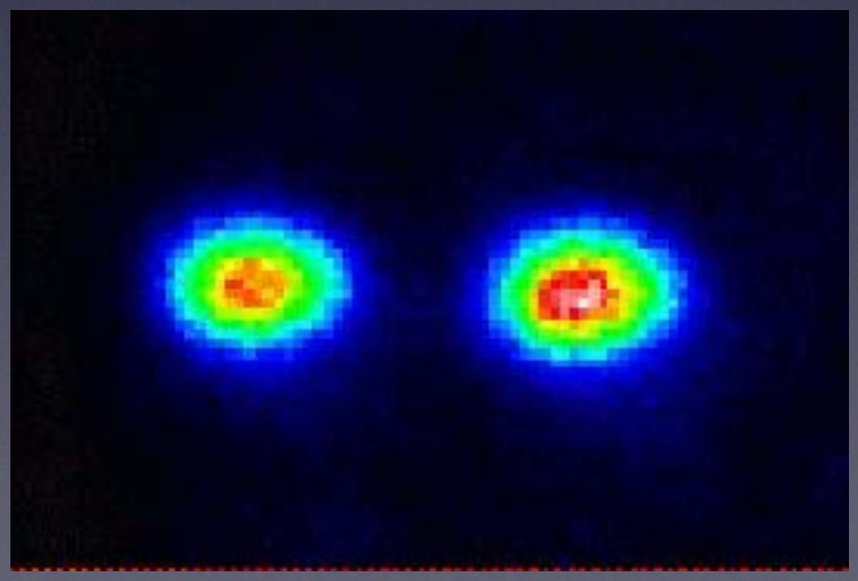


Feynman

Quantum Computers

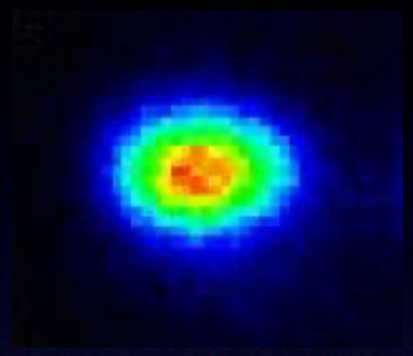


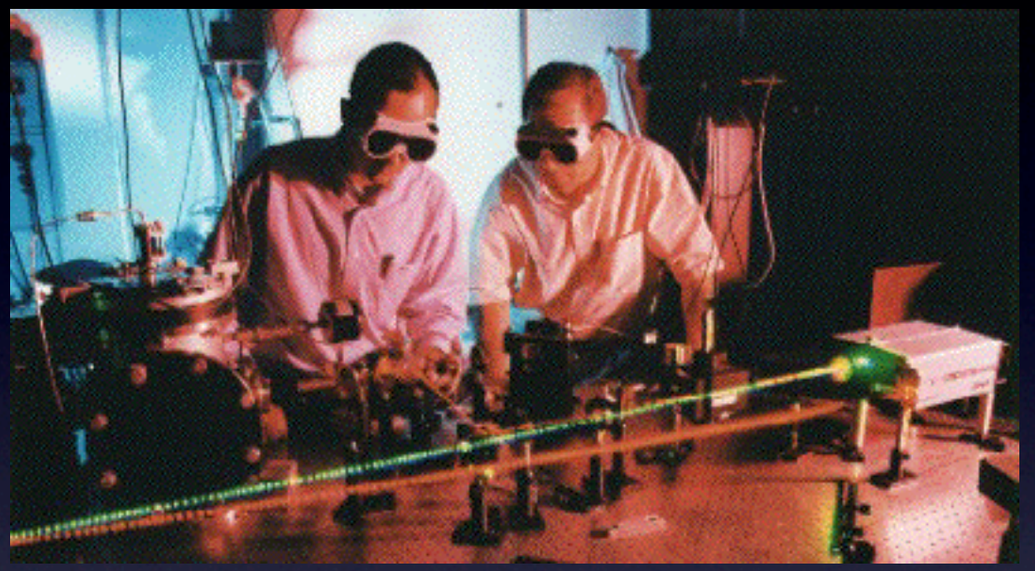
like charges repel

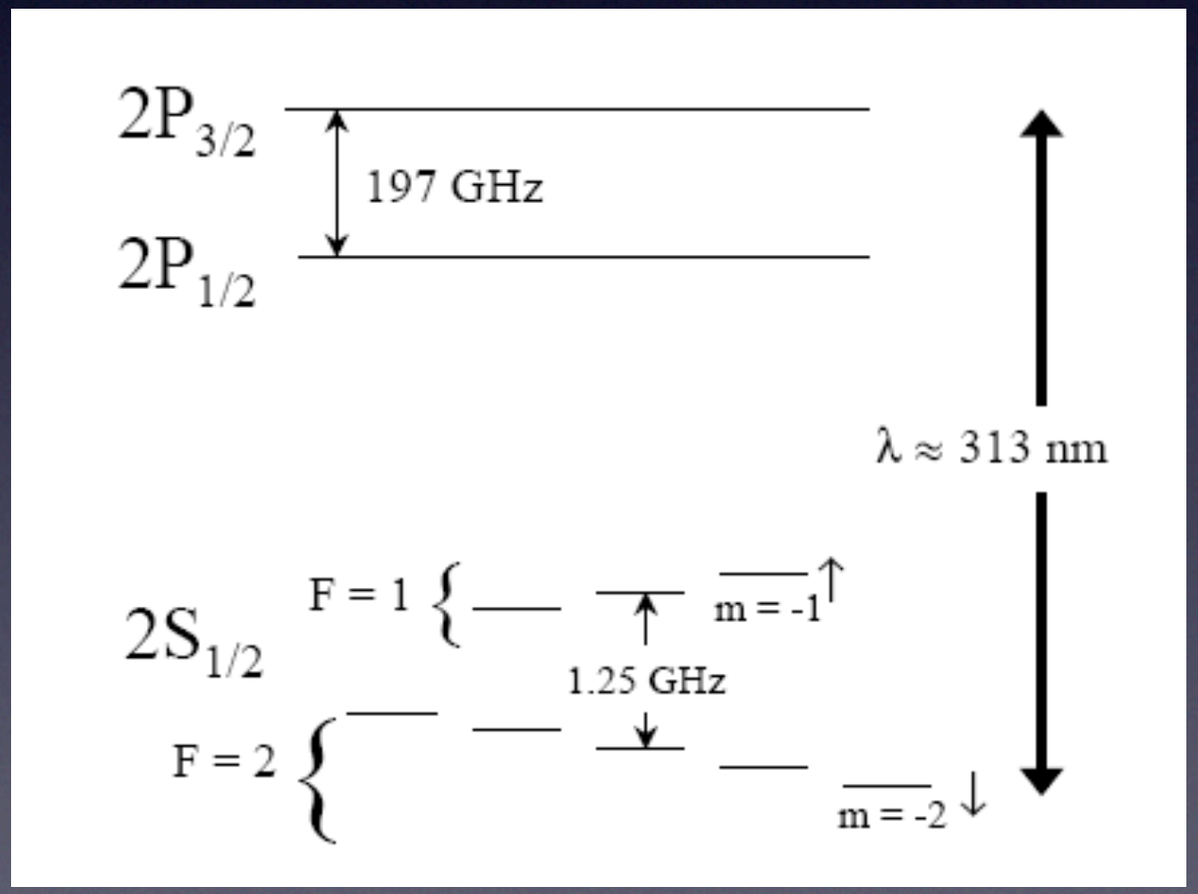


2 $^{9}Be^{+}$ Ions
in an Ion Trap

Quantum Bit (Qubit)







Probe quantum effects by shining in a laser with precisely tuned frequency

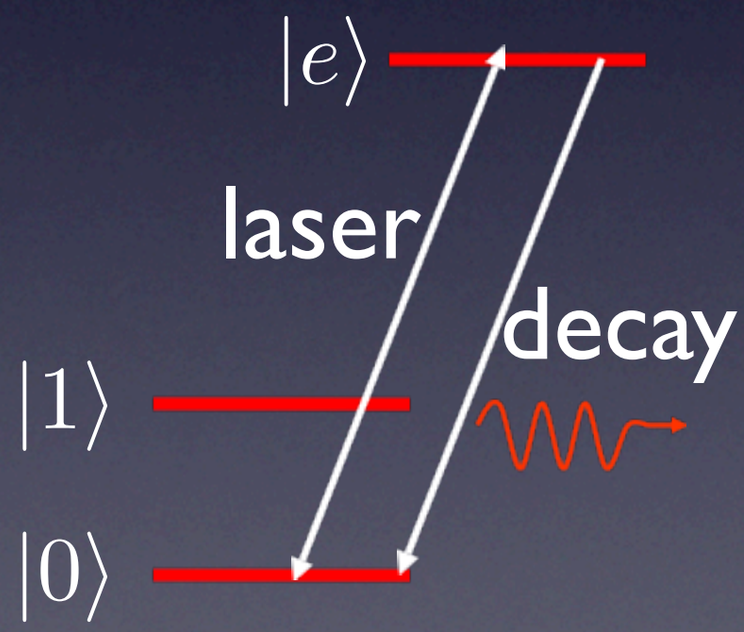
Qubit

"Prepare"



Laser cool to put system in ground state $|0\rangle$

"Measure"



If system is in $|0\rangle$ lots of light will be emitted

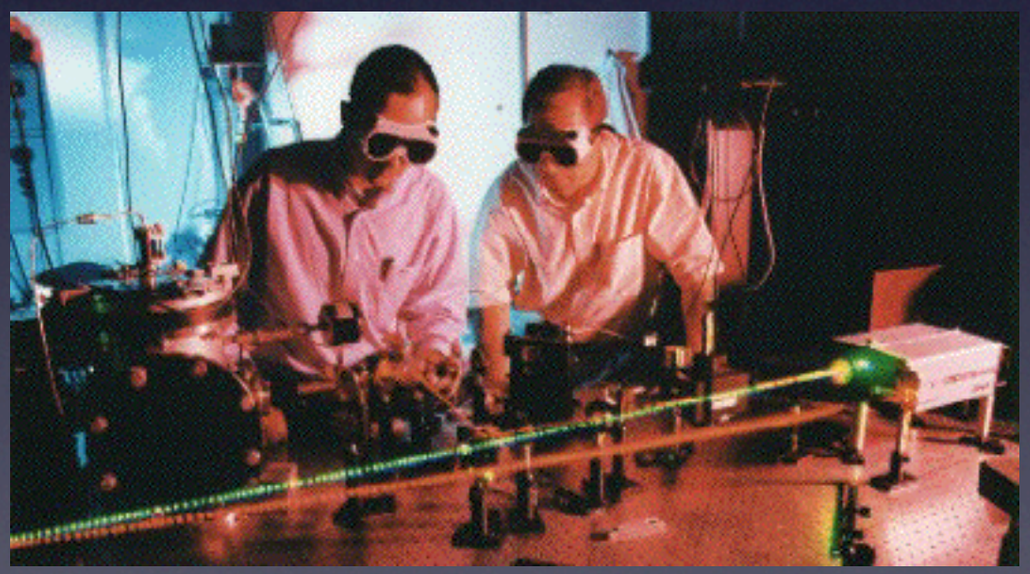
If system is in $|1\rangle$ no light will be emitted

Qubit

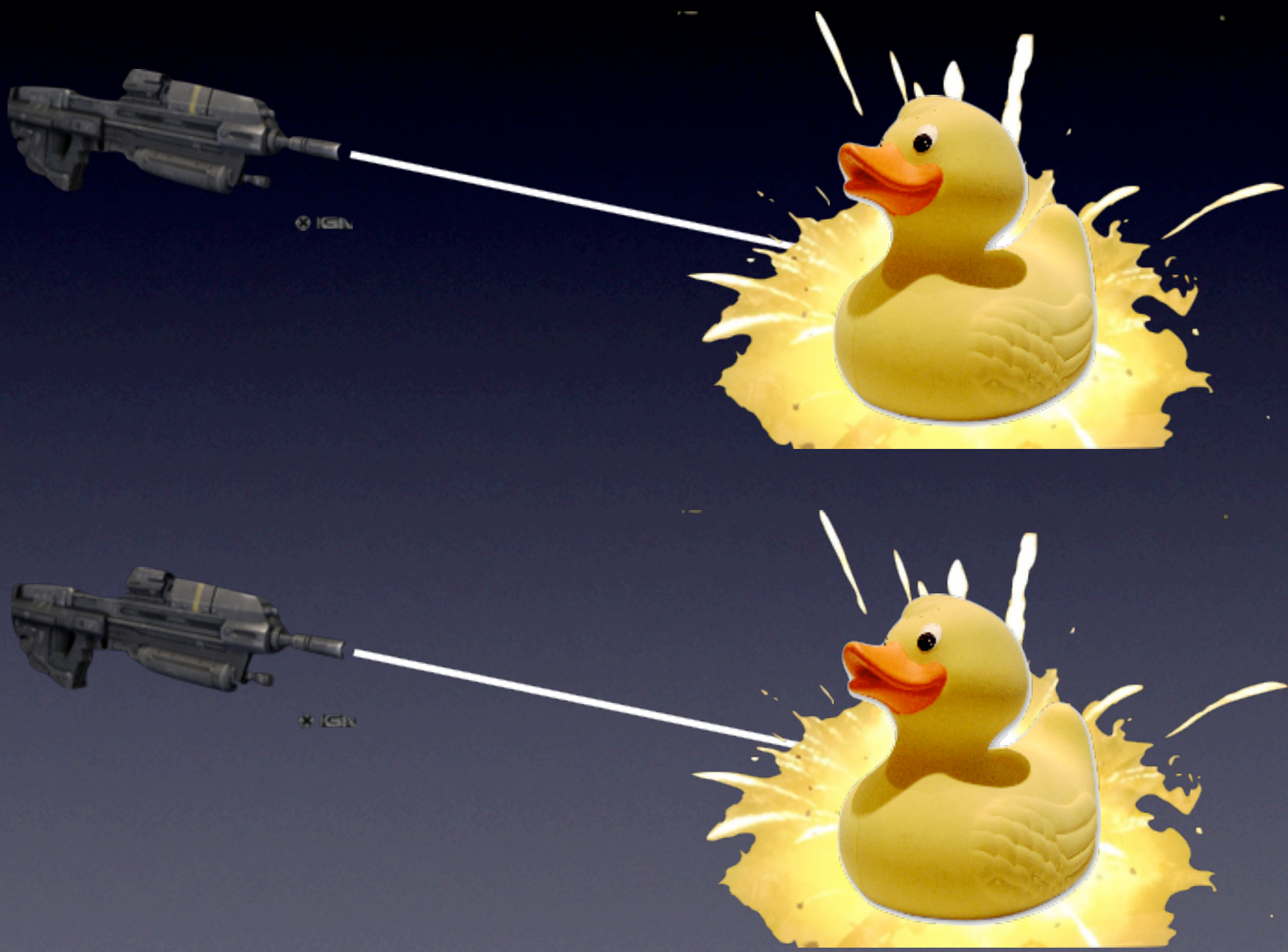
"Manipulate"



apply a laser to change the state

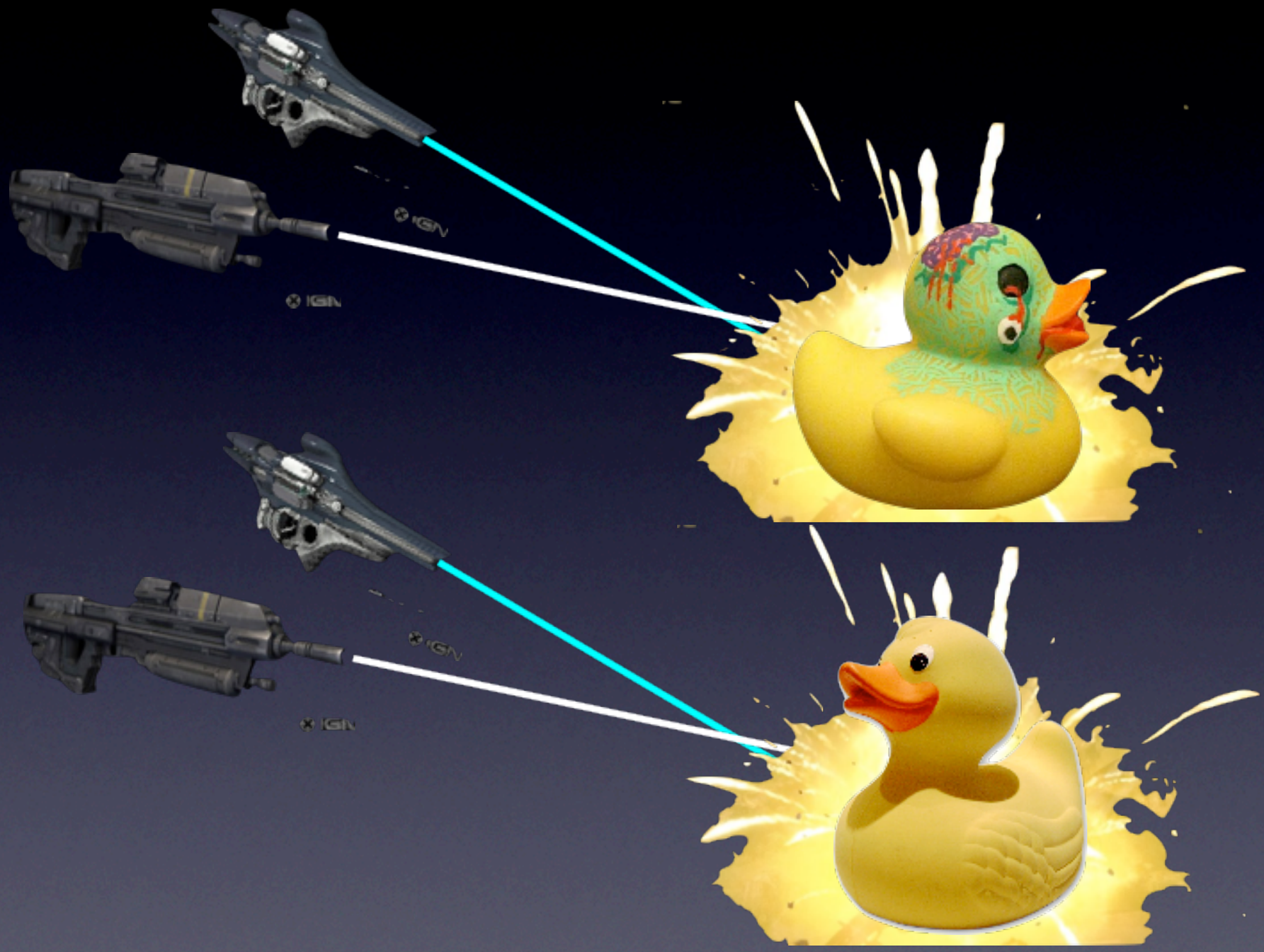


Quantum “Video Game”



Box = Duck

Plasma Rifle!

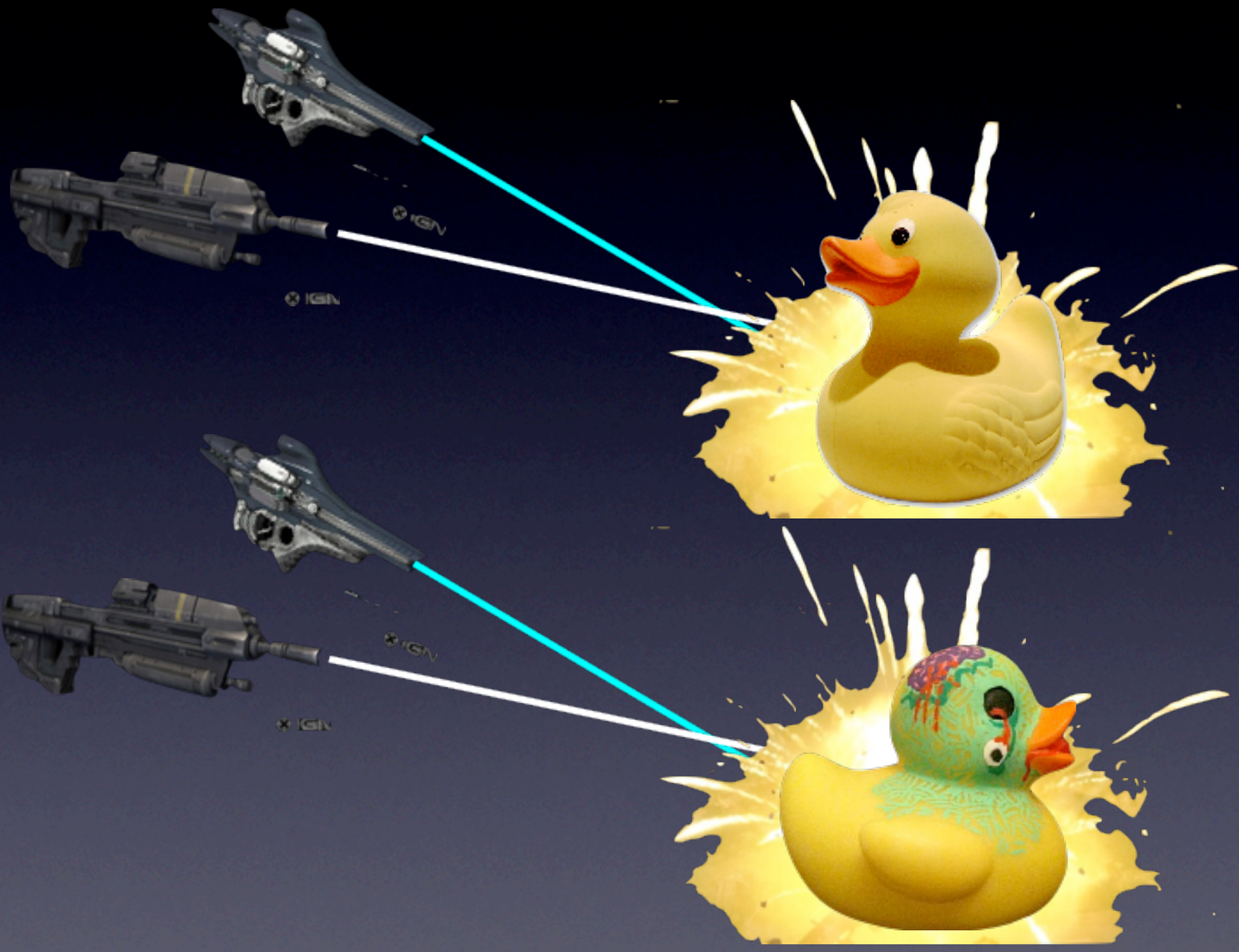


50%

50%

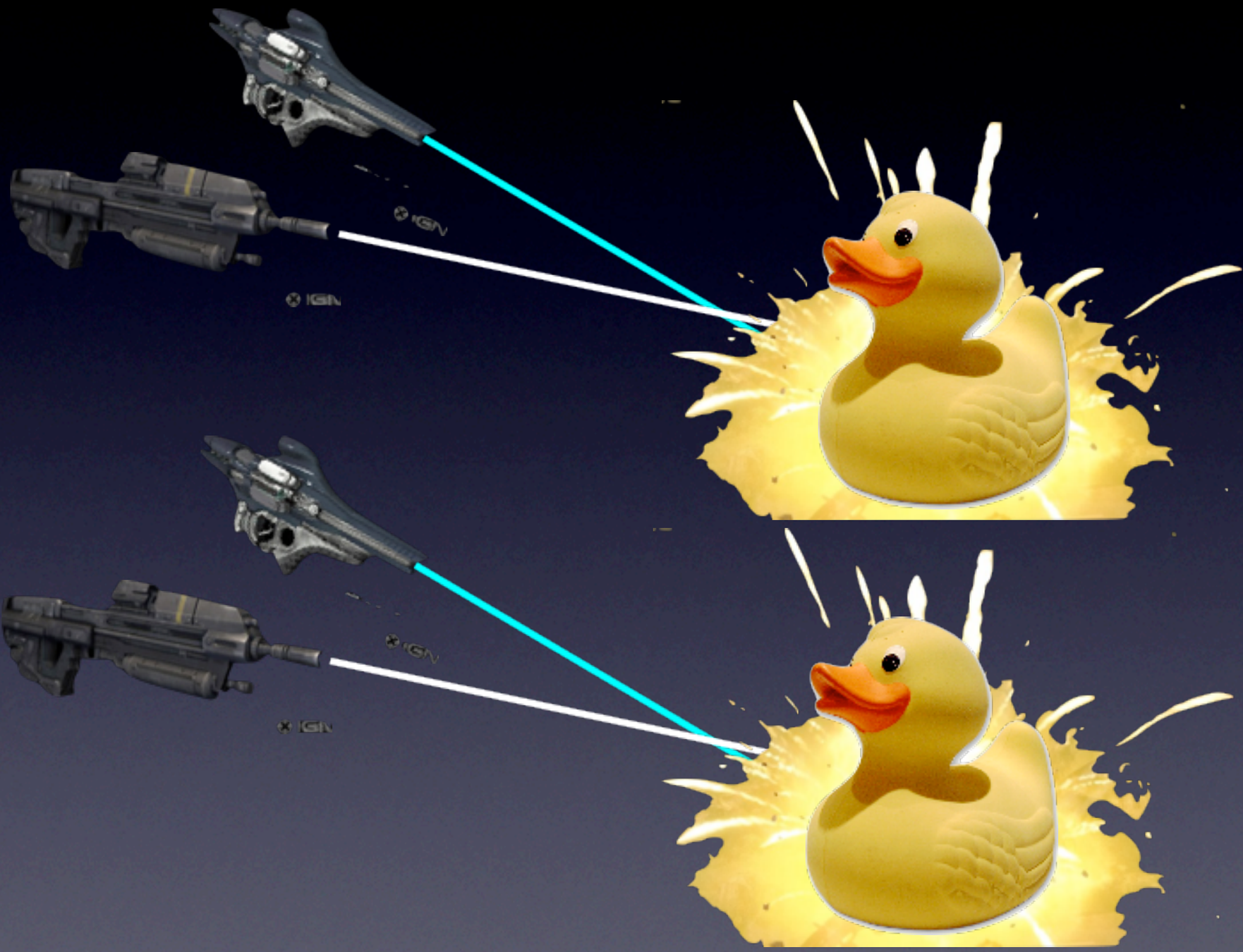
Hypothesis: Plasma rifle kills duck 50% of the time

Duck, Duck



50%

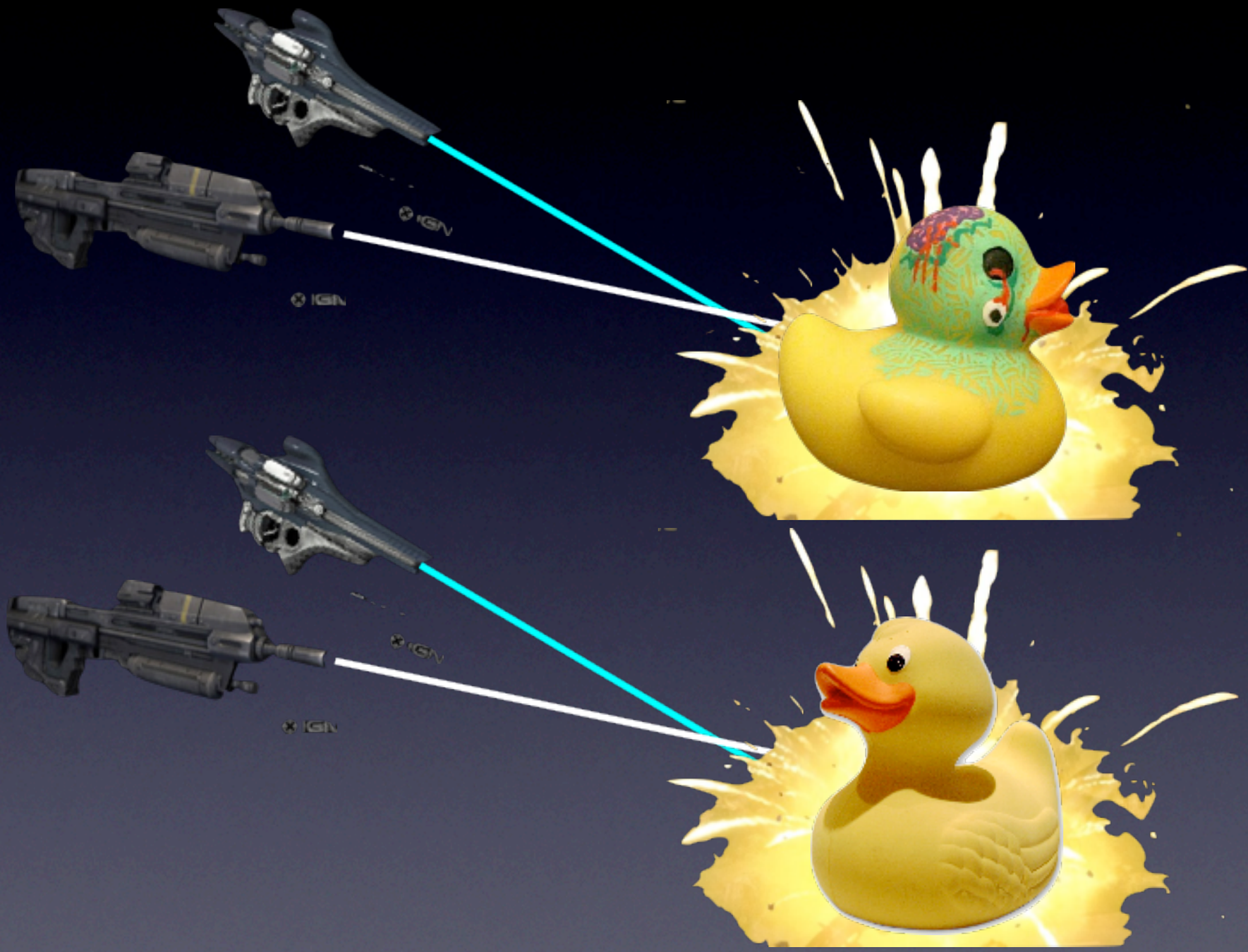
Duck, Duck



50%

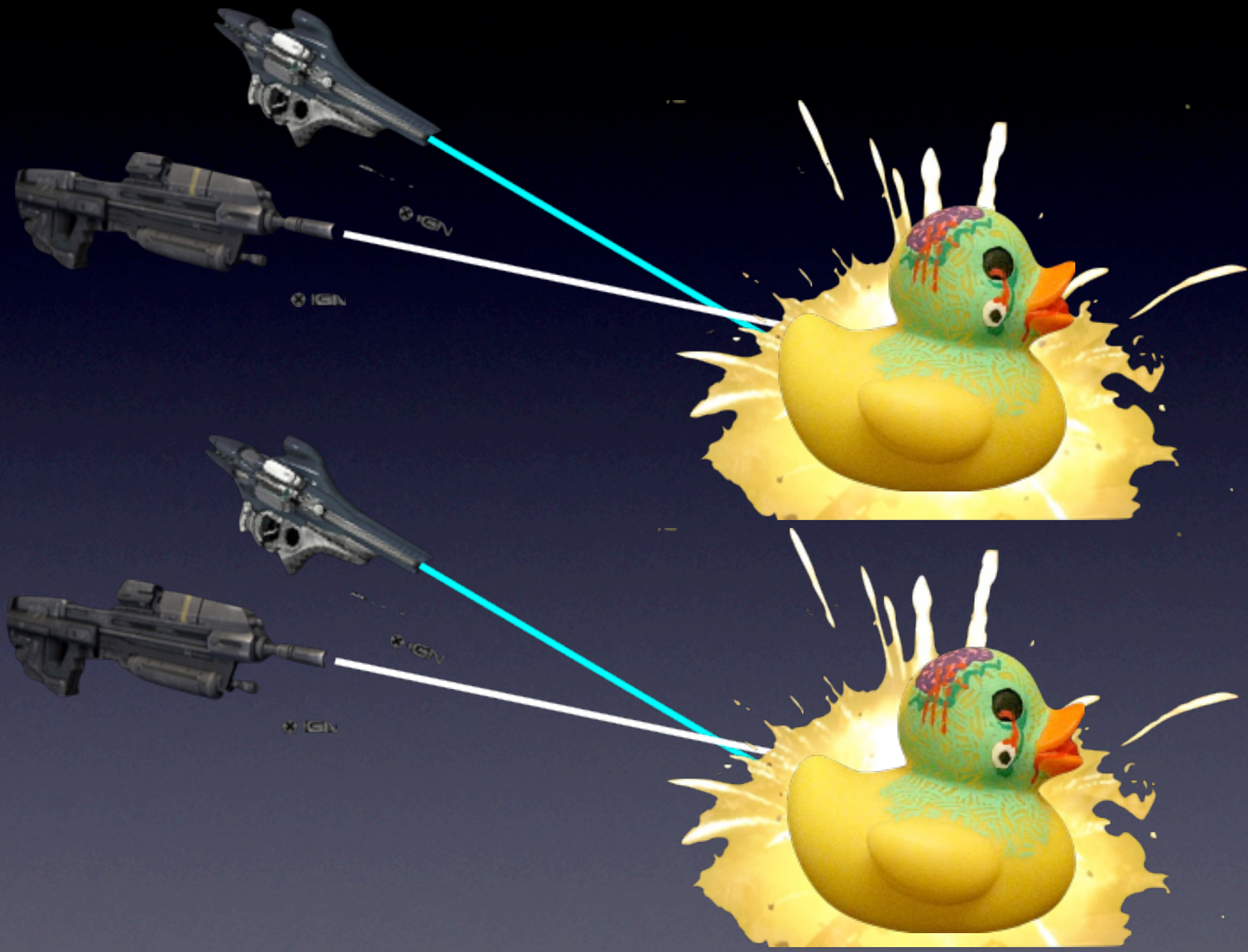
Hypothesis: Plasma rifle kills duck 50% of the time

Dead Duck, Duck



50%

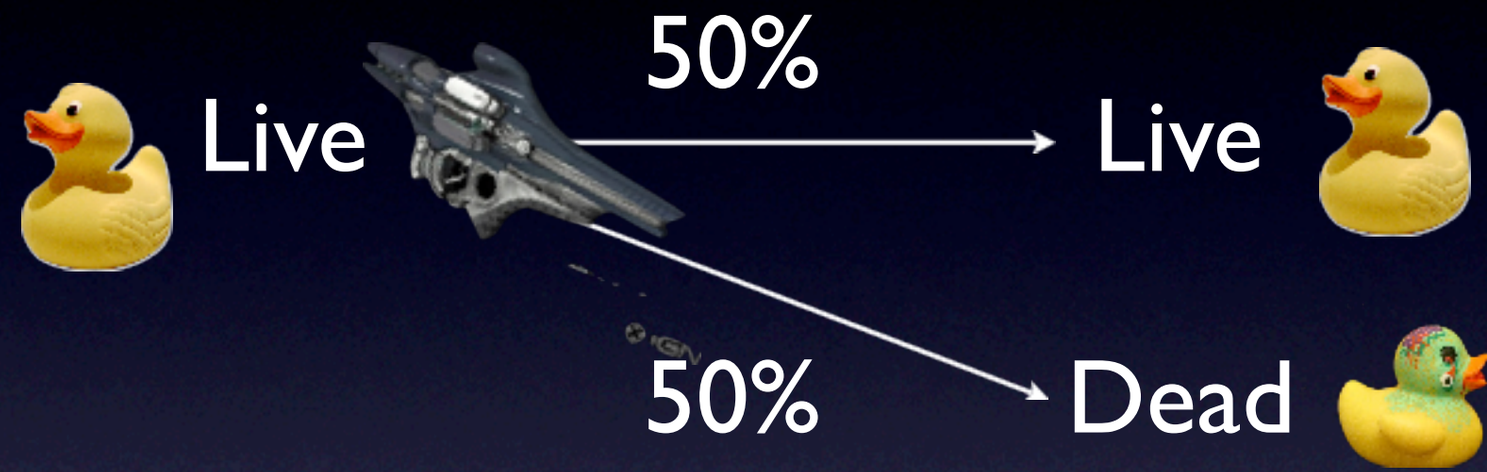
Dead Duck, Dead Duck

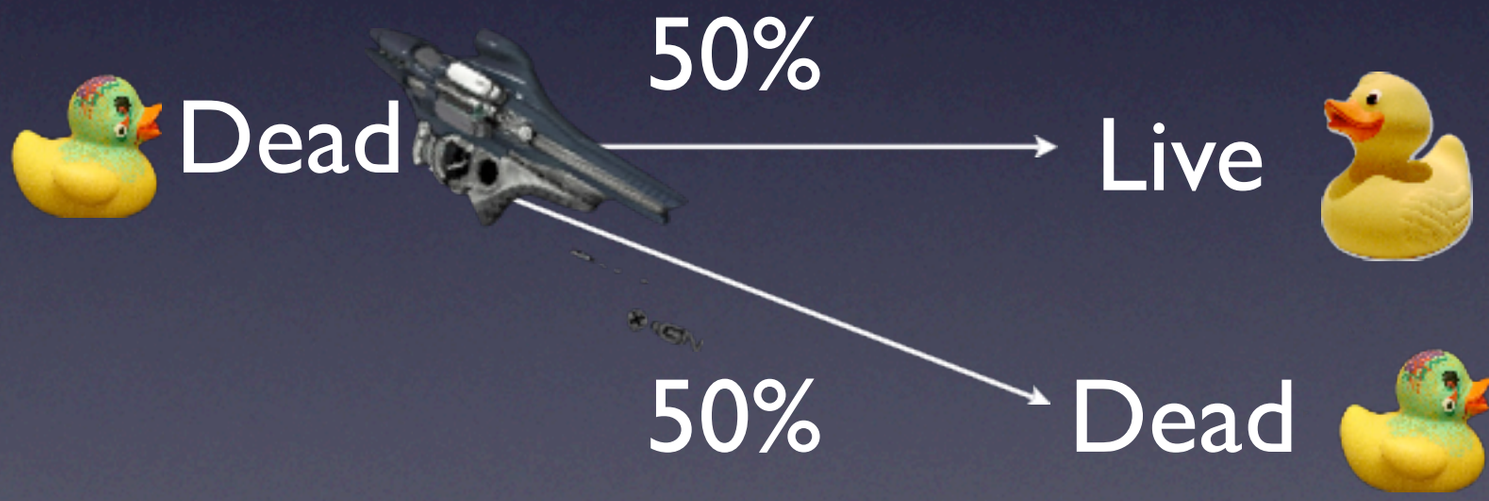


50%

Hypothesis: Plasma rifle resurrects duck 50% of the time

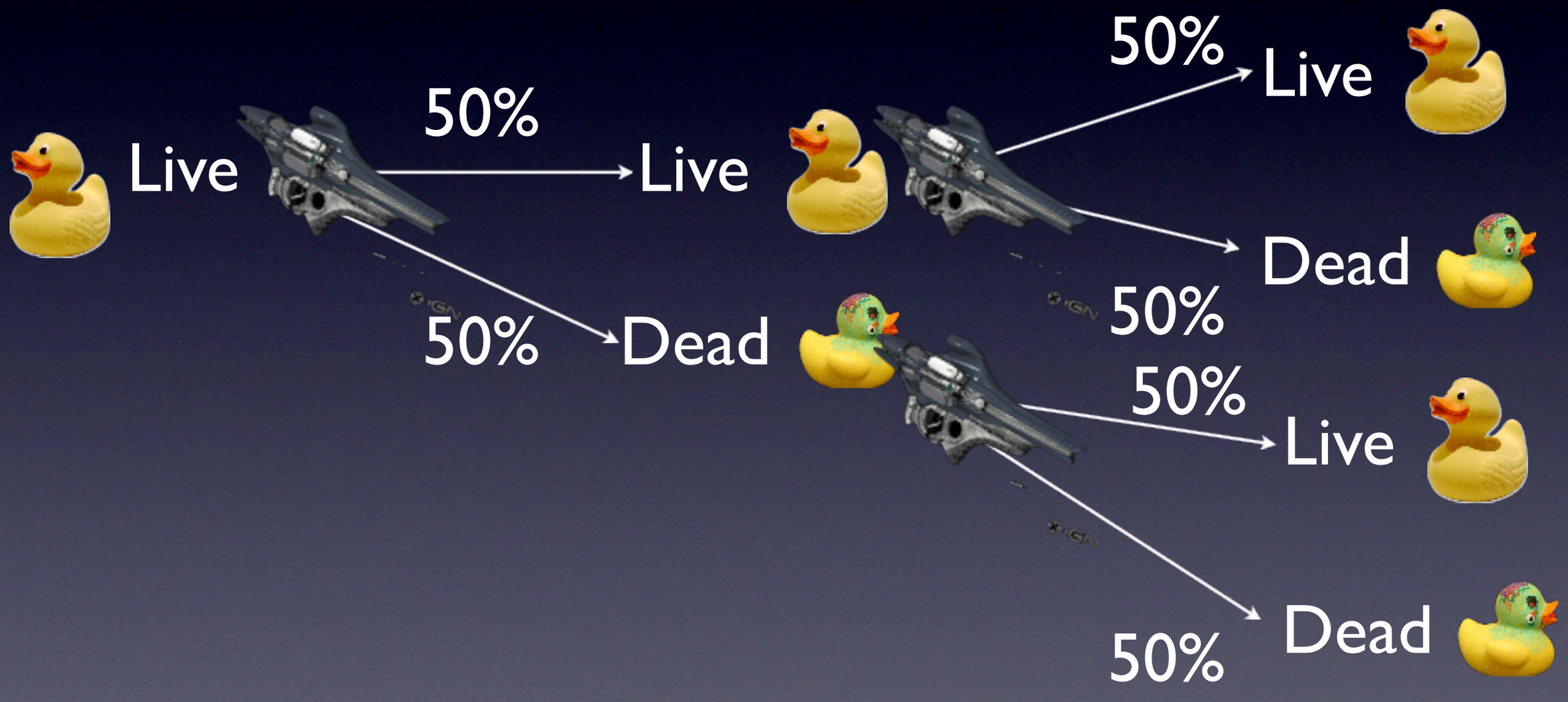
Review





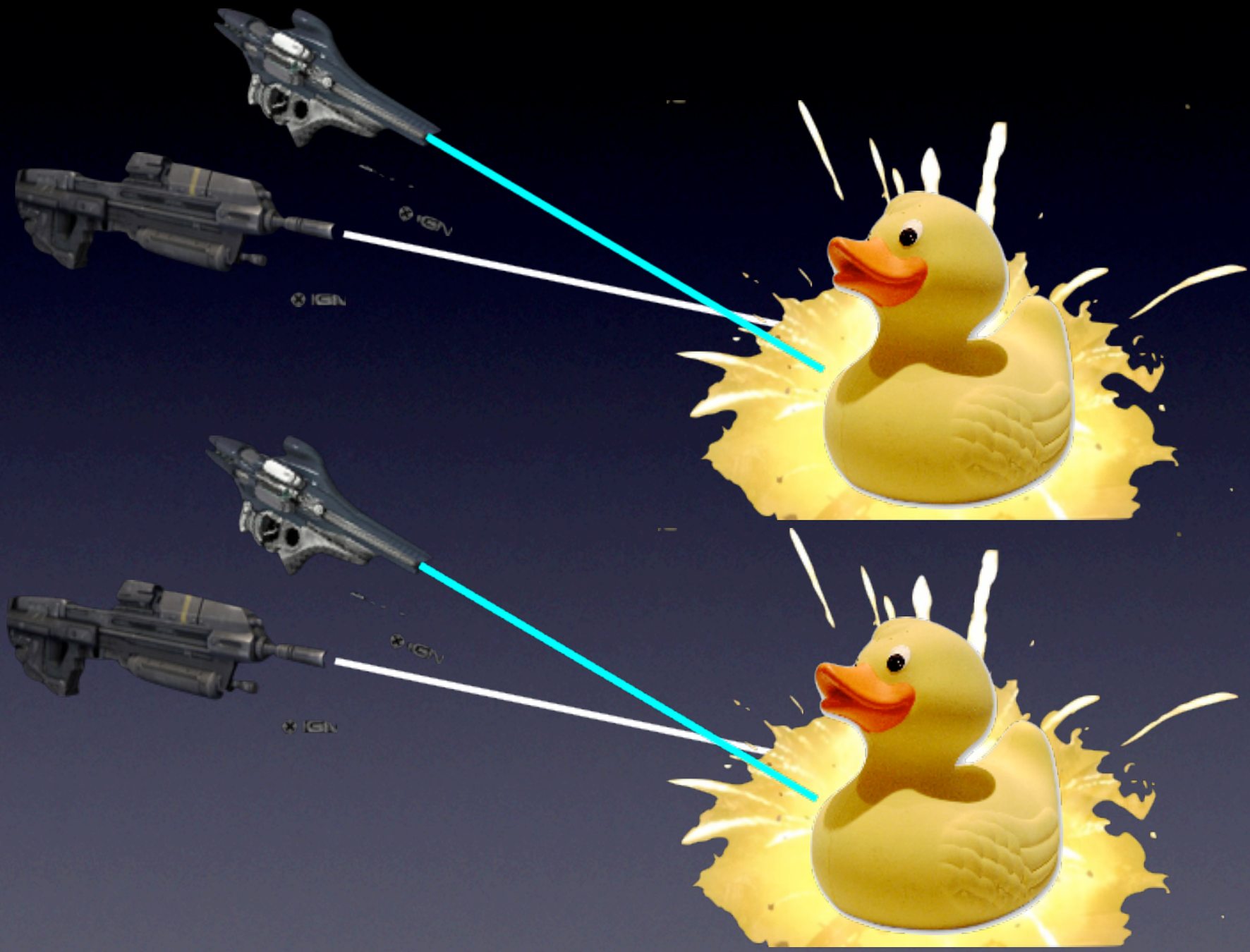
Question

Start with live duck, shoot it two times with plasma riffle, open box, what appears?



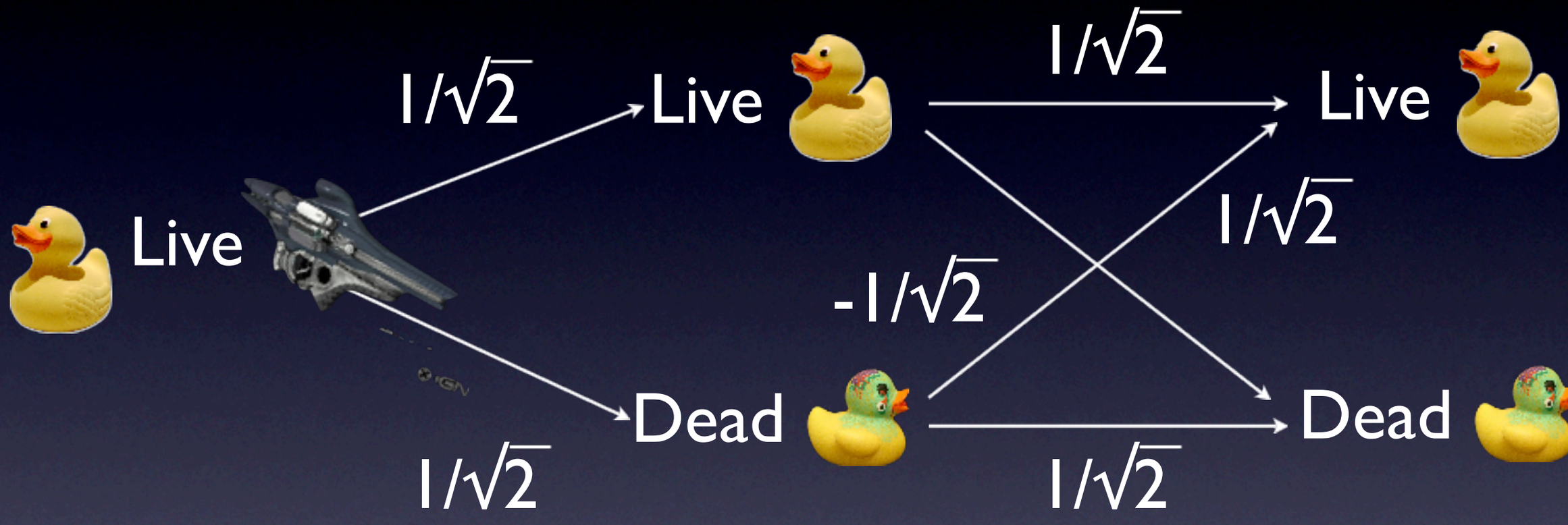
Answer: 50 % live + 50 % dead

Plasma Rifle x 2



Huh?

Answer



amplitudes

1/√2 Live
1/√2 Dead

square them

1/2 Live
1/2 Live

1 Live ⇒ 1 Live

square them

Quantum Video Games



Wrong!

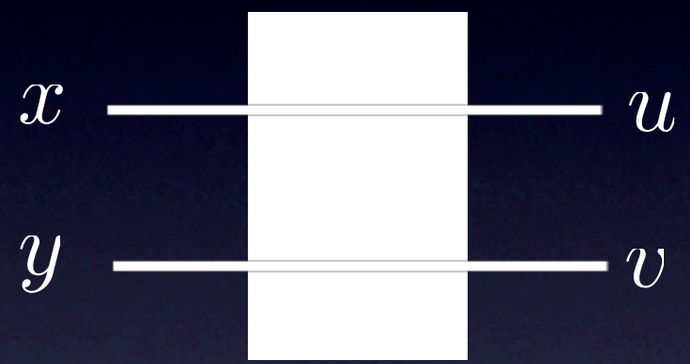
Deutsch's Problem



David Deutsch

Dr. Falcon





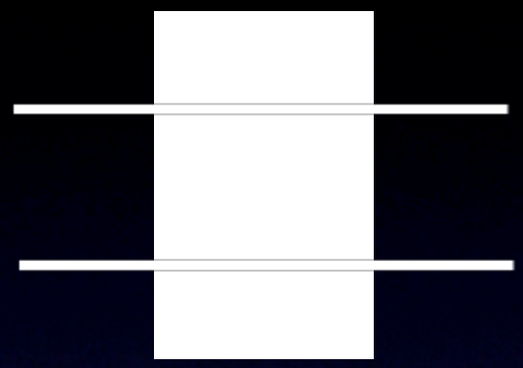
$xy \to uv$	$xy \to uv$	$xy \to uv$	$xy \to uv$
$00 \to 00$	$00 \to 01$	$00 \to 00$	$00 \to 01$
$01 \to 01$	$01 \to 00$	$01 \to 01$	$01 \to 00$
$10 \to 10$	$10 \to 11$	$10 \to 11$	$10 \to 10$
$11 \to 11$	$11 \to 10$	$11 \to 10$	$11 \to 11$

constant

balanced

Deutsch's Problem

$$\begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix}$$



$00 \to 00$ $00 \to 01$ | $00 \to 01$ $00 \to 00$
$01 \to 01$ $01 \to 00$ | $01 \to 00$ $01 \to 01$
$10 \to 10$ $10 \to 11$ | $10 \to 10$ $10 \to 11$
$11 \to 11$ $11 \to 10$ | $11 \to 11$ $11 \to 10$

constant

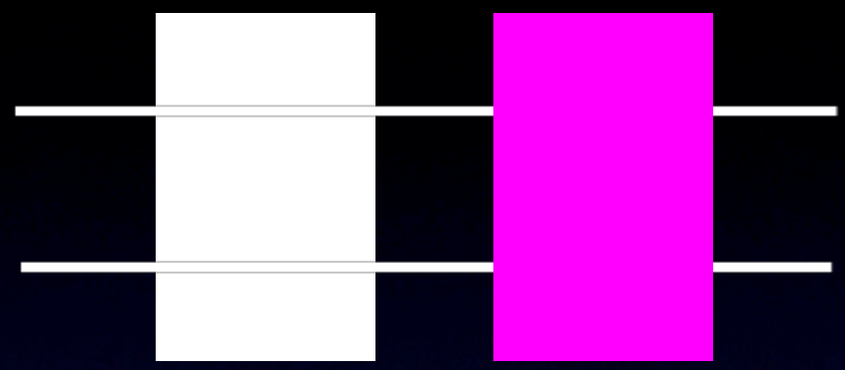
$$\begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \quad \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

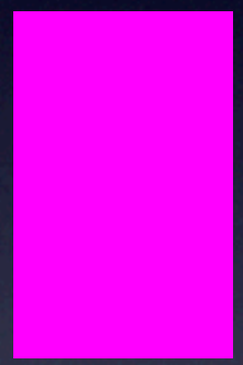
balanced

$$\begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \quad \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

Deutsch's Problem

$$\begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix}$$





$$= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

constant

$$\begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \end{bmatrix}$$

01

balanced

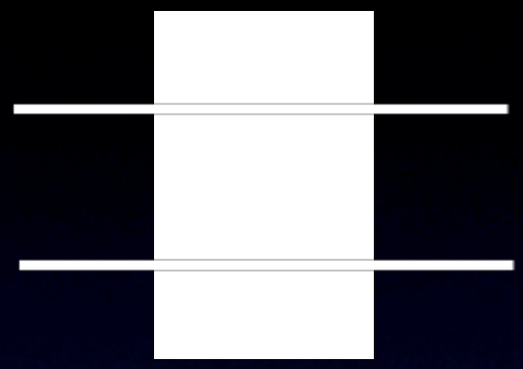
$$\begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \rightarrow \begin{bmatrix} 0 \\ 0 \\ 0 \\ -1 \end{bmatrix}$$

11

Deutsch's Problem



1985



- Classically: 2 queries
- Quatumly: 1 query

Might the computational complexity depend on physics?

Beyond a Curiosity



Quantum computers offer superpolynomial query complexity speedups over classical computers [Bernstein, Vazirani 1993]

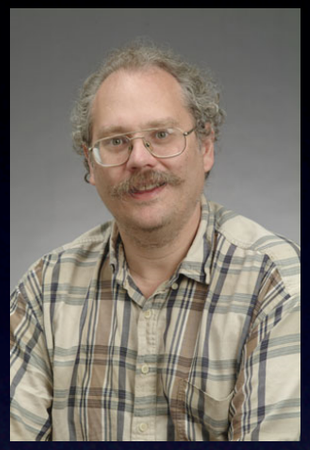
Umesh
Vazirani
(Berkeley)

Quantum computers offer exponential query complexity speedups over classical computers [Simon 1994]



Dan Simon
(Microsoft)

Shor's Breakthrough



Peter Shor

1881988129206079638386972394616504398071635633794173827007633564229888597152346654853190606065047430453173880113033967161996923212057340318795506569962213051687593076502570599807508642406493739712550055038649119906436234252670840638518957594638895726176858331794727721461074353025362230719730482246329146953020971164598521711305207112563635903975275

X

Best classical algorithm

$$O\left(\exp(n^{\frac{1}{3}})\right)$$

Shor's quantum algorithm

$$O\left(n^3\right)$$

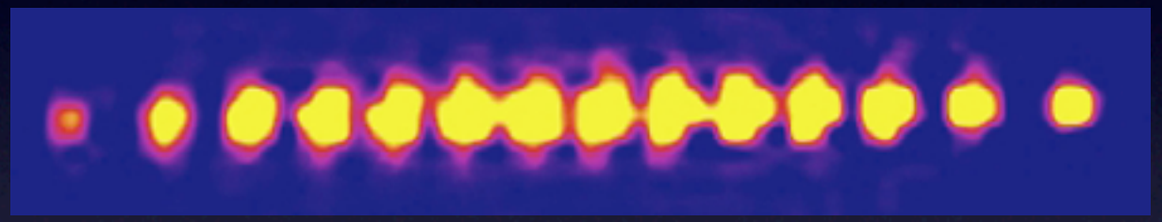
Hyperbole

Only time will tell if and when the problems of building a quantum computer can be overcome….As information becomes the world's most valuable commodity, the economic, political and military fate of nations will depend on the strength of ciphers. Consequently, the development of a fully operational quantum computer would imperil our personal privacy, destroy electronic commerce and demolish the concept of national security. A quantum computer would jeopardize the stability of the world. Whichever country gets there first will have the ability to monitor the communications of its citizens, read the minds of its commercial rivals and eavesdrop on the plans of its enemies. Although it is still in its infancy, quantum computing presents a potential threat to the individual, to international business and to global security

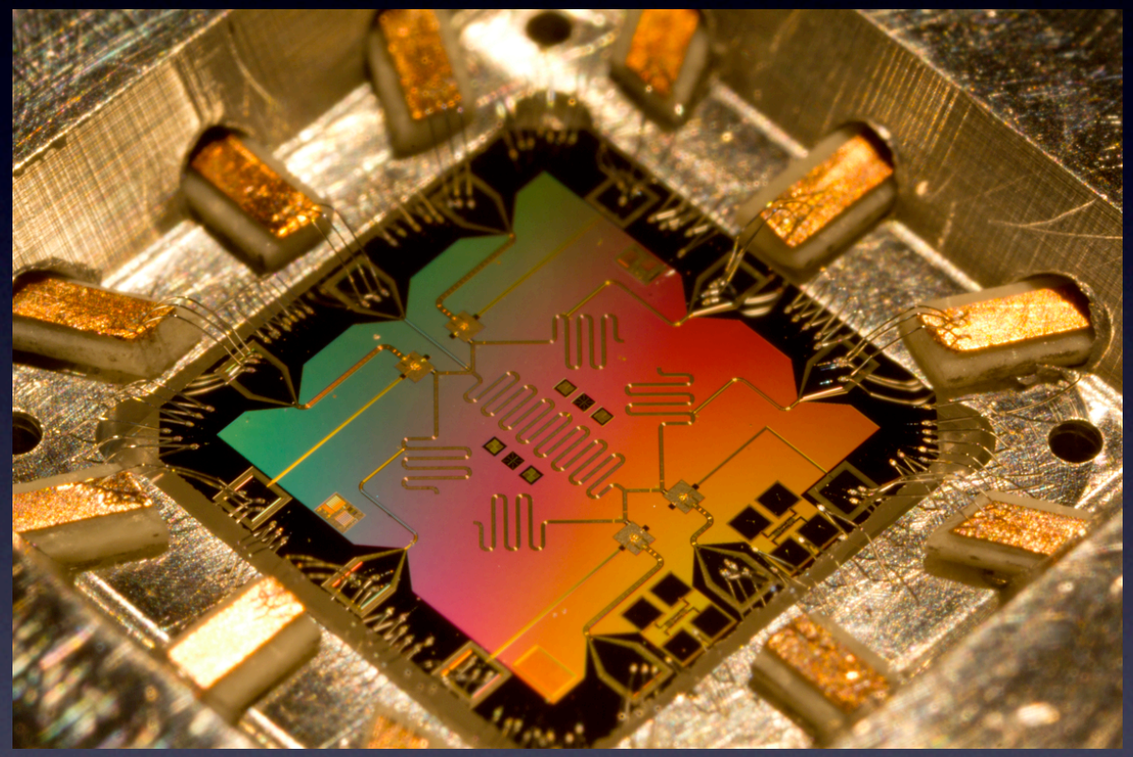


- Simon Singh

Where Are Quantum Computers?



14 qubit ion trap
(Innsbruck)



4 qubit (or 9) using
superconducting circuits
(UCSB)

A Game







Alice **Bob** **Charlie**

An Evil Game







Alice **Bob** **Charlie**

1. Each given slip **X** or **Y**

2. At midnight each party shouts “+1” or “-1”

Winning the Game







Alice	Bob	Charlie	Winning Product
X	X	X	+1
X	Y	Y	-1
Y	Y	X	-1
Y	X	Y	-1

A Game







Alice Bob Charlie

A_X B_X C_X

A_Y B_Y C_Y

All either +1 or -1

A Game







Alice	Bob	Charlie	Product
X	X	X	$A_X B_X C_X$
X	Y	Y	$A_X B_Y C_Y$
Y	Y	X	$A_Y B_Y C_X$
Y	X	Y	$A_Y B_X C_Y$

Winners?

Assuming winning strategy:

$A_X B_X C_X = +1$

$A_X B_Y C_Y = -1$

$A_Y B_Y C_X = -1$

$A_Y B_X C_Y = -1$

Multiply equations….contradiction!

LHS:

$A_X^2 A_Y^2 B_x^2 B_Y^2 C_x^2 C_Y^2 = +1$

Losers



Alice



Bob



Charlie

There is no strategy to win the game :(

In the Real World



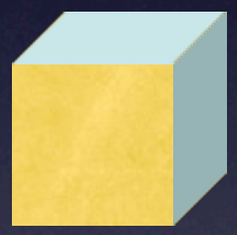
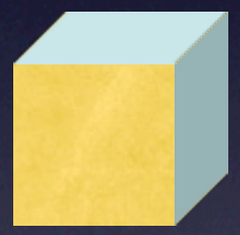
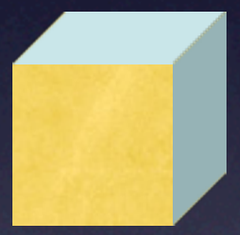
Alice



Bob



Charlie

This is wrong, there is a way to win

Quantum Theory

In the Real World



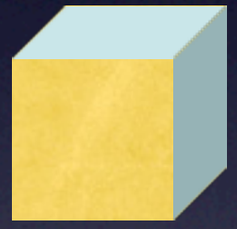
Alice

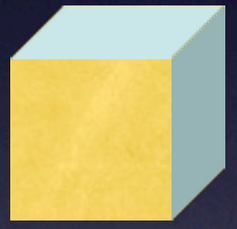


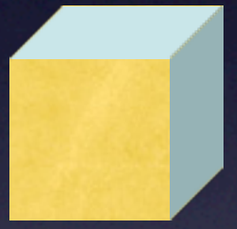
Bob



Charlie







Share $\begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$

Apply $M_X = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$

$M_Y = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$

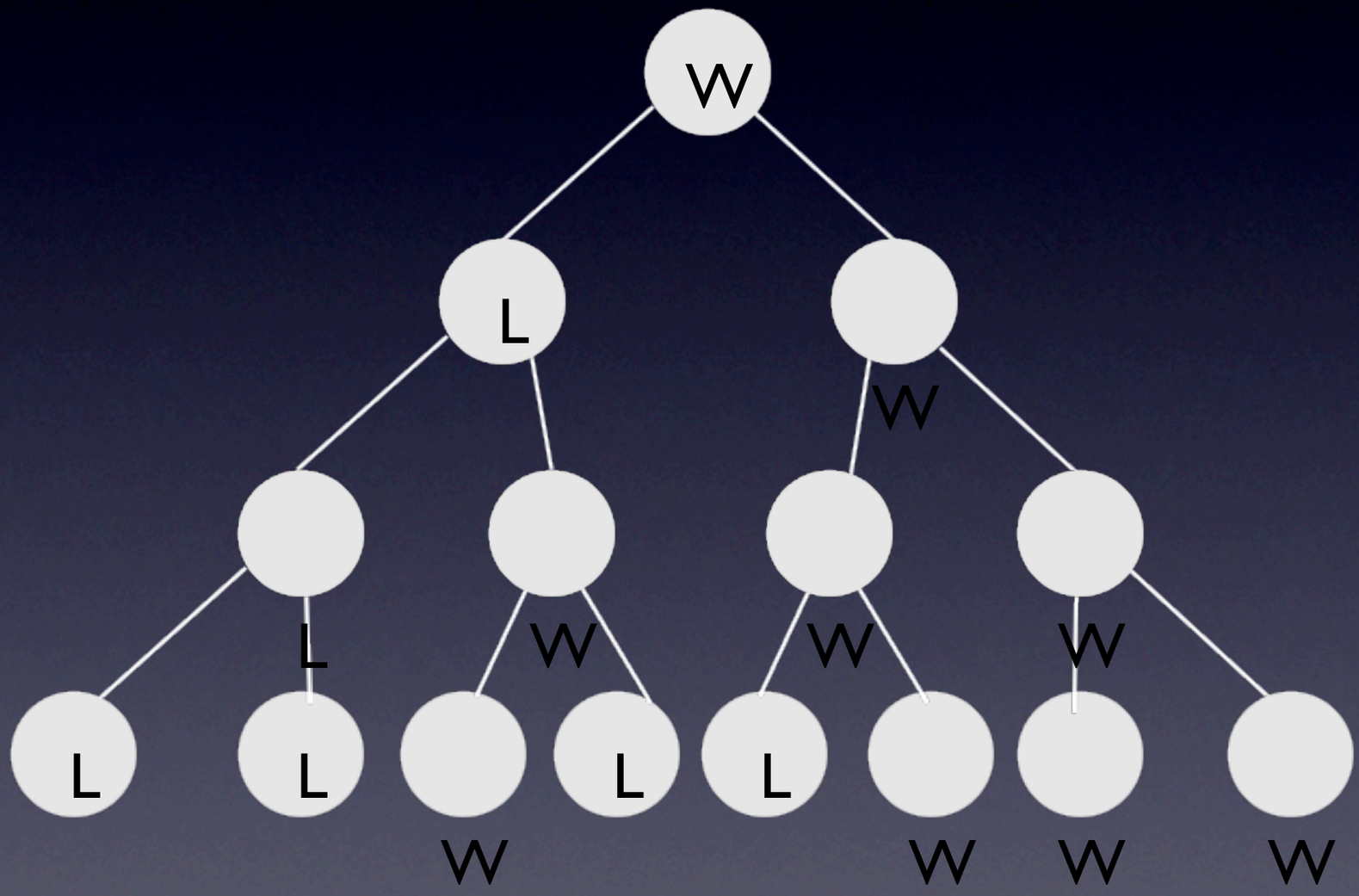
Measure

Game Tree

player 1

player 2

player 1

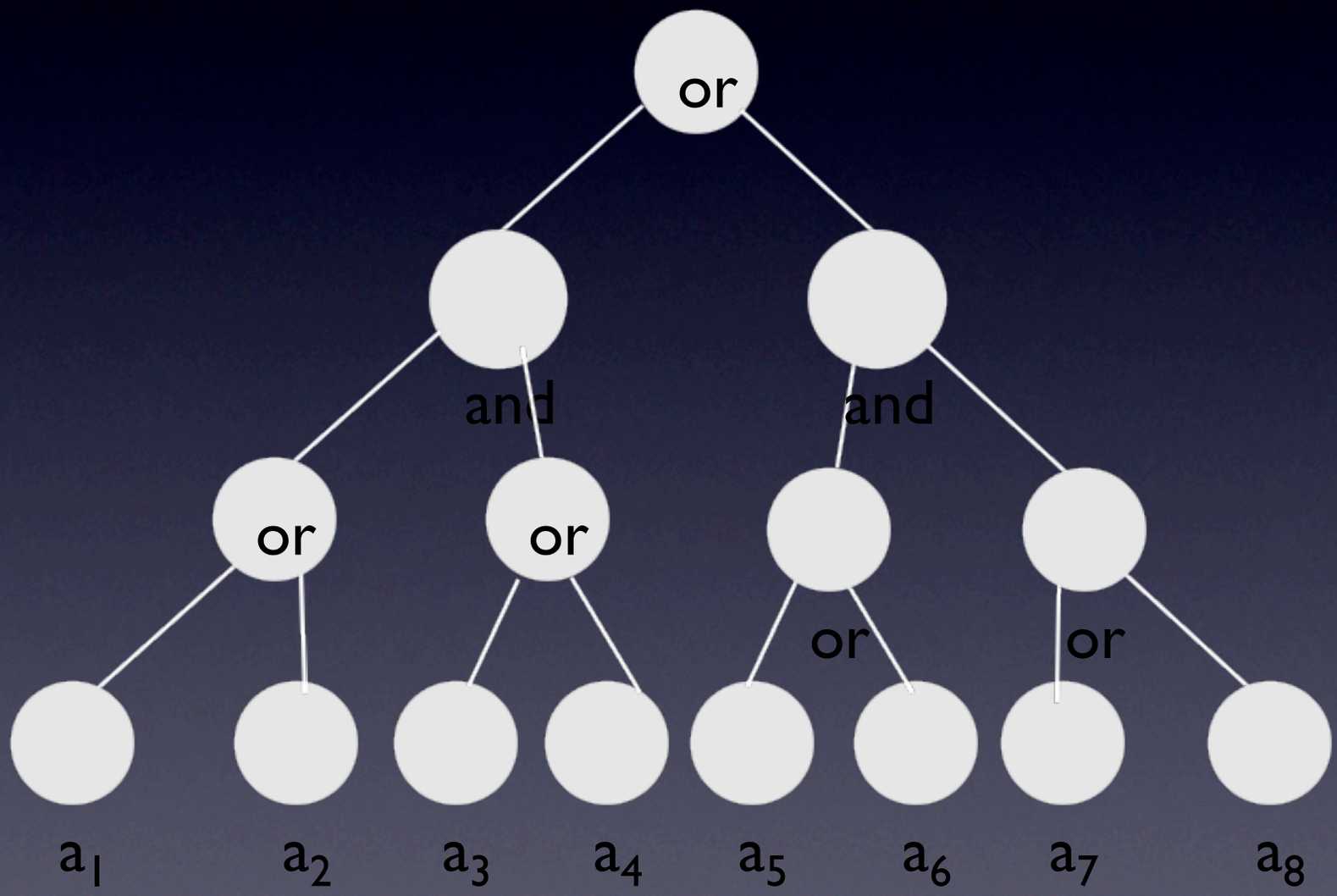


Games

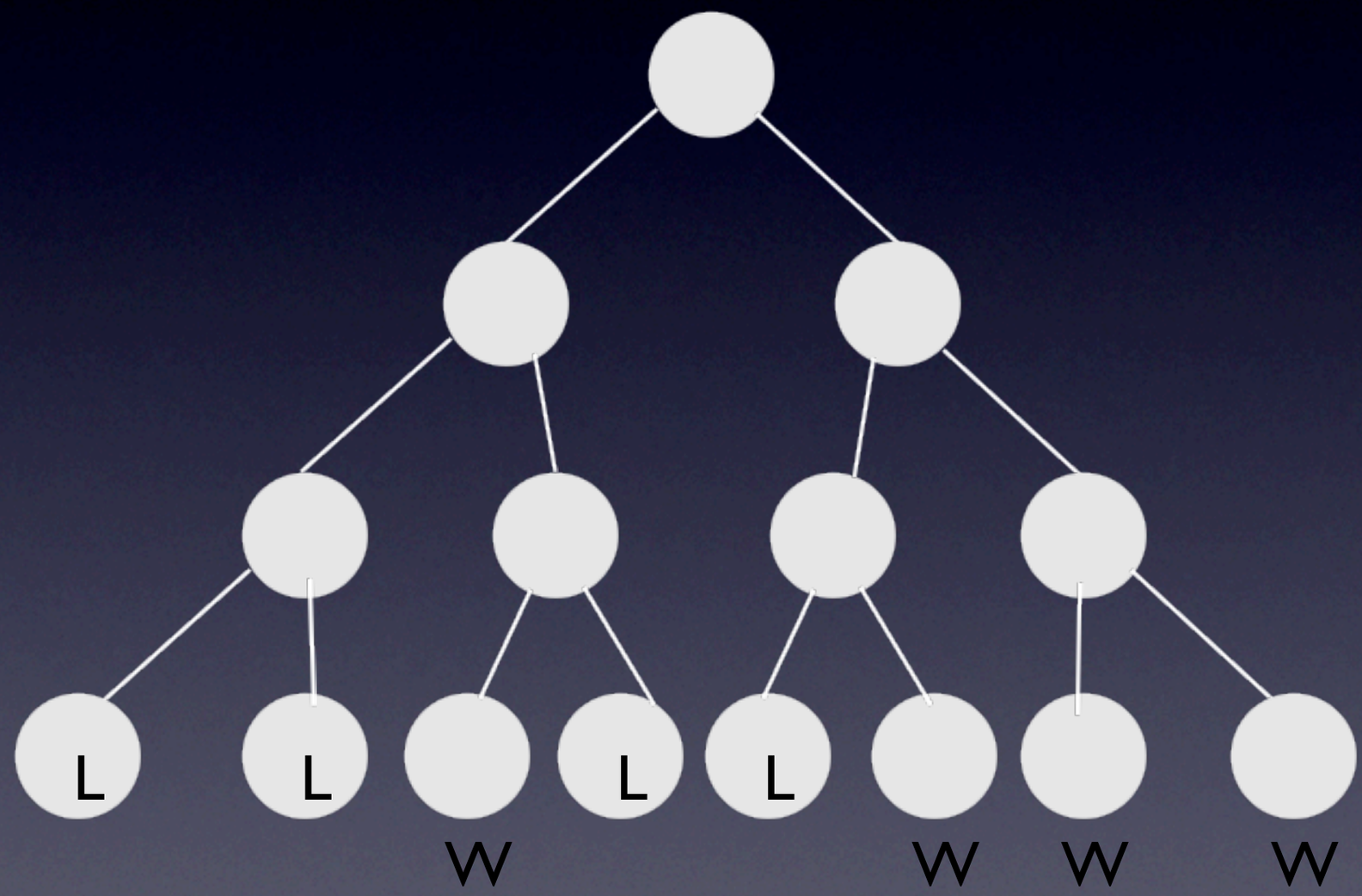
player 1

player 2

player 1

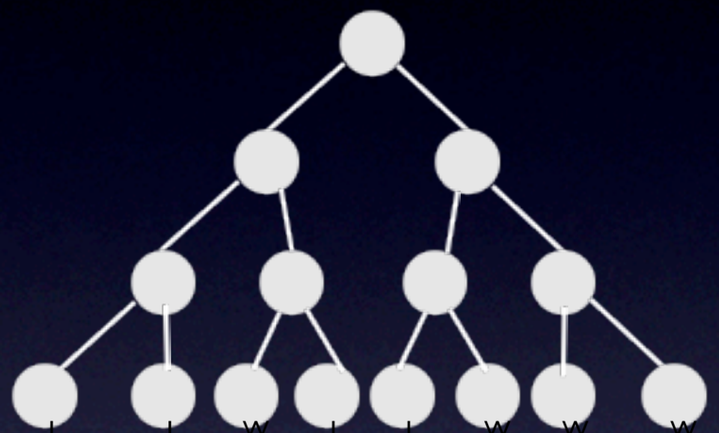


AND-OR Tree Evaluation



How many leaves do you need to query to compute value of root?

AND-OR Tree Evaluation



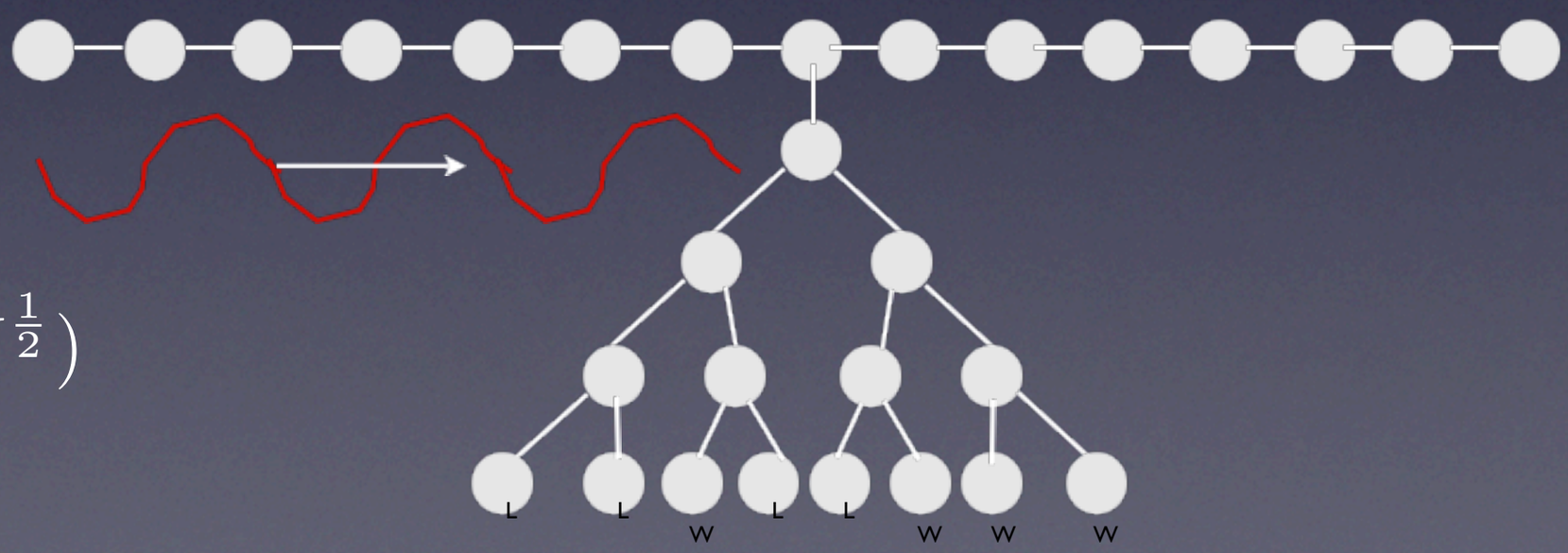
Classically (alpha-beta pruning)

$\Theta(N^{\alpha})$

$\alpha = \log_2(1+\sqrt{33}) - 2$

$\alpha \approx 0.75$

Quantum [Farhi, Goldstone, Gutman 2007]



$\Theta(N^{\frac{1}{2}})$

Video Games on Quantum Computers?????

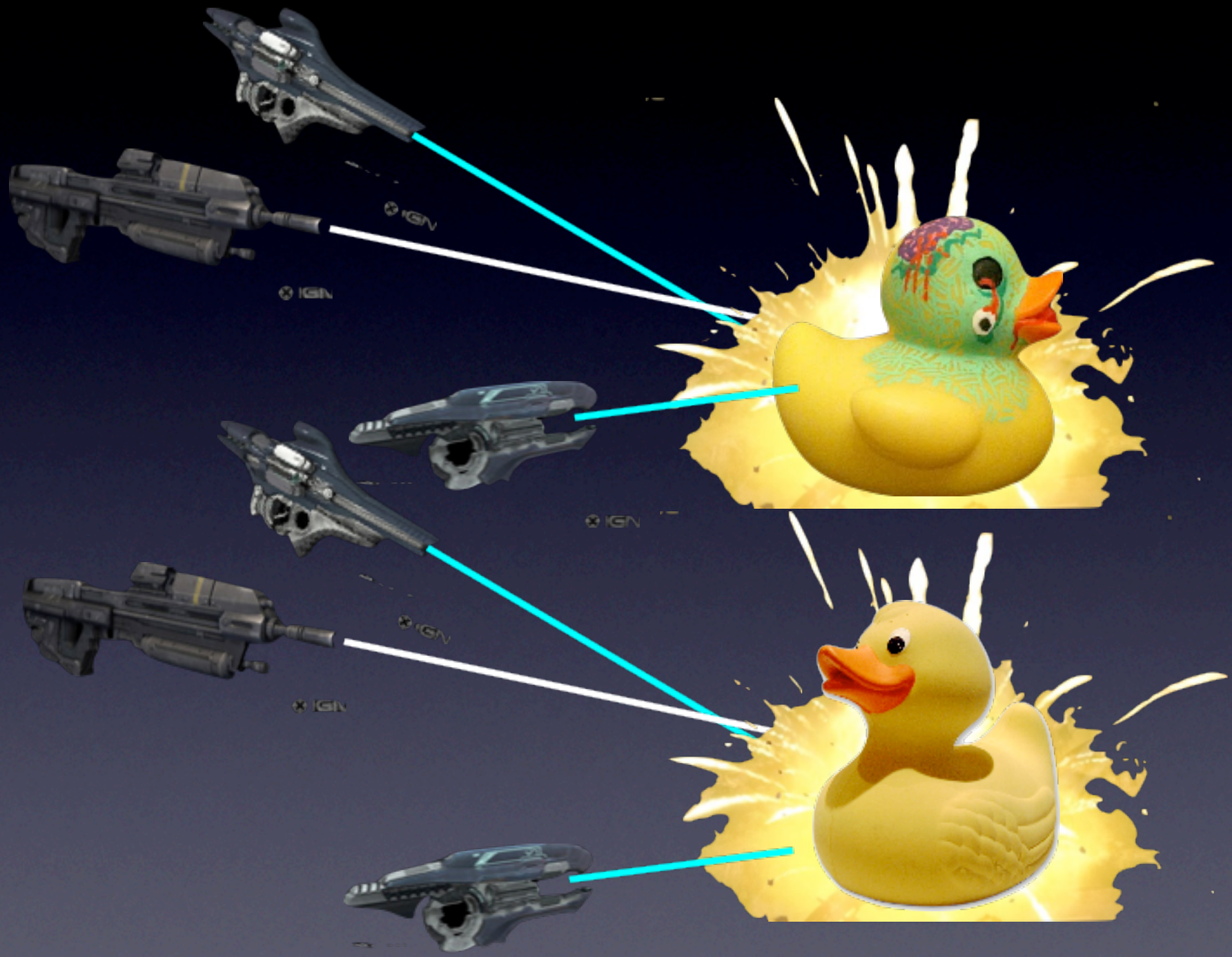


- The world works in a strange way (quantum theory)
- Computers that exploit this can run faster (and break RSA)
- Online games that can't be beat classically would fall if players had entangled qubits
- But it will be a while before we have a large quantum computer... [no need to update your business plan, yet]

The Test Dog



QVG



50%

50%

Hypothesis: Plasma riffle kills duck 50% of the time





